Fueter regular theta fields and formulations of the Riemann hypothesis

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Abstract

We construct an entire left Fueter regular quaternion field from the first spectral jet of the theta heat family, with Newman time retained as an independent real parameter. Starting with the theta Mellin transform, we derive the normalization relating this family to the Riemann xi function. We state the classical global heat-flow inputs and prove the equivalence between the Riemann hypothesis and nonvanishing of the quaternion field at every real argument and positive time. A translation identity gives an equivalent zero-free half-plane for one fixed quaternion field. Two stationary Laguerre signs, a locally bounded loss formulation, and a scalar vertical-convexity criterion are derived and translated with explicit coordinate and multiplication conventions. Cauchy–Fueter reproduction yields signed boundary correlations; harmonic currents and norm identities retain their precise limitations. For the complete theta source we prove an explicit norm gap on $0 \le t \le 1$, $|x| \le 8$, and a quantitative transfer from a full angular gluing defect to four physical jets. Remote perturbation controls and endpoint cancellations identify why absolute estimates and separate theta-index diagonals do not settle the remaining all-height sign problem. A validated compensated calculation extends both stationary signs to $0 \le t \le 1/2$, $7 \le |x| \le 41$; combined with the central theorem it gives an explicit physical norm gap through height 41. Correlated Taylor transport extends the validated norm gap and both signs through height 100. A guarded continuation theorem propagates strict stationary signs forward until branch regularity or boundary control fails. Shifted complete-source contours give absolute jet errors at the gamma exponential scale for arbitrary heights; the remaining signed target is localized to a moving small-time region. No proof of the Riemann hypothesis is claimed.

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1 Scope and conventions

The Riemann hypothesis concerns the zeros of a particular scalar entire function. Quaternion regularity is a property of its constrained lift; the arithmetic content remains the complete theta source. The central equivalence proved here is

$$\mathrm{RH} \quad \Longleftrightarrow \quad F(t,x) \neq 0 \quad \text{for all } t > 0,\ x \in \mathbb{R},$$

where F is the explicit field constructed below. This is pointwise nonvanishing. It requires neither a positive lower bound uniform in all heights nor a simplicity assertion at time zero.

Hamilton units satisfy $\mathbf{i}^2 = \mathbf{j}^2 = \mathbf{k}^2 = \mathbf{ijk} = -1$ and $\mathbf{ij} = \mathbf{k}$. The complex field $\mathbb{C}_i = \mathbb{R} + \mathbf{i}\mathbb{R}$ is identified with the usual complex numbers. Quaternion components are written on the right of $\mathbf{j}$: $Q = A + \mathbf{j}B$, $A, B \in \mathbb{C}_i$. In particular $\mathbf{j}A = \overline{A}\mathbf{j}$. Quaternion conjugation is denoted by a bar; when its argument is complex it is ordinary complex conjugation. The real part of a quaternion is $\operatorname{Re} Q$, its imaginary quaternion part is $\operatorname{Im} Q$, and $|Q|^2 = \overline{Q}Q$.

We keep the following roles distinct:

Symbol	Role
$t \in \mathbb{R}$	independent Newman time
$z = x + iy \in \mathbb{C}_i$	scalar spectral variable
$q = z_0 + \mathbf{j}z_1 \in \mathbb{H}$	quaternion argument with independent complex coordinates
$\tau = t + \overline{z_1}$	complex heat-time displacement inside the lift
$h_m = \partial_z^m H$	scalar spectral derivative of order m
$\mathscr{L}_\ell = h_{\ell+1}^2 - h_\ell h_{\ell+2}$	Laguerre form, $\ell = 0, 1$
$n, m \geq 1$	theta-series indices when a double sum is displayed

Derivative-order indices may be written j or ℓ; the quaternion unit $\mathbf{j}$ is always bold. All quaternion differential operators multiply units on the left unless a right action is stated. A prime on a scalar heat slice differentiates its spectral argument, never its time parameter. An angular theta argument used later will be called y_{ang} when a distinction is needed. The known threshold and localization theorems are stated as classical inputs; the deductions from them and every coordinate translation used here are proved in the text. The theta estimates have the explicitly bounded domains displayed in their statements.

2 The scalar theta family and its normalization

The quaternion construction begins with a fixed scalar family. We derive its normalization so that subsequent changes of coordinates cannot change the meaning of the Riemann hypothesis. Write

$$\vartheta(y) = \sum_{n\in\mathbb{Z}} e^{-\pi n^2 y}, \qquad \vartheta_+(y) = \sum_{n\geq 1} e^{-\pi n^2 y}, \qquad y > 0.$$

The Gaussian Fourier transform and Poisson summation give

$$\vartheta(y) = y^{-1/2}\vartheta(1/y). \tag{2.1}$$

Consequently the balanced real-variable field

$$P(u) = e^u\vartheta(e^{4u}), \qquad A(u) = e^u\vartheta_+(e^{4u}), \qquad P(u) = e^u + 2A(u)$$

is even. These series and their derivatives converge locally uniformly on the real line, and in a complex neighborhood of every real point. Differentiating the series gives

$$\begin{aligned}\Phi(u) &= \frac{P''(u) - P(u)}{16} = \frac{A''(u) - A(u)}{8} \\ &= \sum_{n\geq 1} \left(2\pi^2 n^4 e^{9u} - 3\pi n^2 e^{5u}\right) e^{-\pi n^2 e^{4u}}.\end{aligned} \tag{2.2}$$

Thus the full real source is even and real analytic; in particular all its odd derivatives at zero vanish. For $u \geq 0$, each summand is positive, since its prefactor is $\pi n^2 e^{5u}(2\pi n^2 e^{4u} - 3)$. Also

$$0 < \Phi(u) < 4\pi^2 e^{9u - \pi e^{4u}}, \qquad u \geq 0. \tag{2.3}$$

Indeed, after dropping the negative term, successive terms of $n^4 e^{-\pi n^2 e^{4u}}$ have ratio at most $16e^{-3\pi} < 1/2$. The geometric sum proves (2.3).

For $\operatorname{Re} s > 1$, termwise integration gives the theta Mellin identity

$$\pi^{-s/2}\Gamma(s/2)\zeta(s) = \int_0^\infty y^{s/2-1}\vartheta_+(y)\,dy.$$

This is also the standard Mellin representation in [1]. Splitting at one and using (2.1), namely $\vartheta_+(y) = (y^{-1/2} - 1)/2 + y^{-1/2}\vartheta_+(1/y)$, gives

$$\xi(s) = \frac{1}{2} + \frac{s(s-1)}{2} \int_1^\infty \left(y^{s/2-1} + y^{(1-s)/2-1} \right) \vartheta_+(y)\, dy, \tag{2.4}$$

where $\xi(s) = s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s)/2$ in its initial domain. The integral on the right is entire in s, providing analytic continuation and $\xi(s) = \xi(1-s)$. The usual completion cancels the trivial zeta zeros and its removable singularities; its zeros are the nontrivial zeta zeros. The Euler product in $\operatorname{Re} s > 1$ and the functional equation place these zeros in $0 \le \operatorname{Re} s \le 1$.

Set $s = 1/2 + iz/2$ in (2.4) and substitute $y = e^{4u}$. Since $s(s-1) = -(1+z^2)/4$, one obtains

$$\xi(1/2 + iz/2) = \frac{1}{2} - (1+z^2) \int_0^\infty A(u) \cos(zu)\, du. \tag{2.5}$$

Evenness of P gives $A'(0) = -1/2$. The field A and its derivatives decay double exponentially at positive infinity. Two integrations by parts therefore yield

$$\int_0^\infty A''(u) \cos(zu)\, du = -A'(0) - z^2 \int_0^\infty A(u) \cos(zu)\, du.$$

Using (2.2) and (2.5) proves the precise normalization

$$H(t,z) := \int_0^\infty e^{tu^2} \Phi(u) \cos(zu)\, du, \qquad H(0,z) = \frac{1}{8}\xi\left(\frac{1}{2} + \frac{iz}{2}\right). \tag{2.6}$$

It agrees with the heat-family convention in [4]. Under this affine change of variable, the critical line is precisely real z, while the critical strip is contained in $|\operatorname{Im} z| \le 1$. Thus RH is equivalent to every zero of $H(0,\cdot)$ being real.

2.1 Analytic properties and the physical heat sign

On a compact subset of complex (t,z), the integrands for every fixed mixed derivative are bounded by a constant times

$$u^N \exp\{Tu^2 + (Z+9)u - \pi e^{4u}\},$$

which is integrable. Differentiation under the integral and local uniform convergence prove joint entireness in (t,z) and

$$\partial_t H = -\partial_z^2 H. \tag{2.7}$$

Increasing Newman time is therefore backward heat in the spectral coordinate. For real t, the function is real entire and even, with Schwarz reflection $H(t,\bar{z}) = \overline{H(t,z)}$. Positivity of the source gives

$$H(t,iy) = \int_0^\infty e^{tu^2} \Phi(u) \cosh(yu)\, du > 0, \qquad t, y \in \mathbb{R}. \tag{2.8}$$

No real-time slice is identically zero, and every spatial zero has finite multiplicity.

Each fixed-time slice has order at most one. To see this directly, absorb $|t|u^2 + 9u$ into $(\pi/2)e^{4u} + C_t$ in (2.3). For $R \ge 1$ the maximum on $|z| \le R$ is bounded by

$$C_t \int_0^\infty e^{Ru - (\pi/2)e^{4u}}\, du \le \frac{C_t}{4} (\pi/2)^{-R/4} \Gamma(R/4).$$

Stirling's estimate consequently gives $\log \max_{|z|\le R} |H(t,z)| \le C_t + CR\log(2+R)$. This growth condition will be used in the real-rooted factorization; it is not dispensable for the necessity of the whole-plane sign criteria.

3 The imported global heat-flow inputs

We state the classical results used below rather than attribute them to quaternion regularity. In the normalization (2.6), de Bruijn's strip contraction says that zeros in $|\operatorname{Im} z| \le a$ at time t_0 lie in $|\operatorname{Im} z| \le \sqrt{\max(a^2 - 2h, 0)}$ at time $t_0 + h$, $h \ge 0$. Newman supplies a finite real threshold Λ such that $H(t, \cdot)$ has only real zeros exactly for $t \ge \Lambda$. These statements are from [2, 3]; the conventions are recorded in [4], Theorem 3.2 and its introduction. The threshold endpoint is included: local uniform convergence and Hurwitz's theorem preserve absence of zeros in every nonreal disk. Consequently

$$\mathrm{RH} \iff \Lambda \le 0, \qquad \Lambda \le \frac{1}{2}. \tag{3.1}$$

Rodgers and Tao prove $\Lambda \ge 0$ [6], so RH is also equivalent to $\Lambda = 0$. Their lower bound is not needed for the collision argument below.

One further, genuinely uniform input is required: there is an absolute $C > 0$ such that for $0 < t \le 1/2$, a zero with $|\operatorname{Re} z| \ge e^{C/t}$ is real. This is [4], Theorem 1.5(i), together with evenness for negative real parts. Separate, uncontrolled localization radii for individual times would not suffice for the compactness argument.

4 A finite collision criterion for RH

Lemma 4.1 (Local Hermite splitting)**.** *Let $W_t = -W_{zz}$ be jointly holomorphic near (T, a), real for real (t, z), and suppose $W(T, \cdot)$ has exact multiplicity $m \ge 2$ at real a. If $\lambda_1, \dots, \lambda_m$ are the roots of the probabilists' Hermite polynomial* He_m*, its nearby zeros satisfy*

$$a + \sqrt{2(t - T)}\,\lambda_k + O(|t - T|), \qquad 1 \le k \le m. \tag{4.1}$$

At least one is nonreal immediately before T.

Proof. Put $W_\ell = \partial_z^\ell W(T, a)$. The heat equation and convergent Taylor expansion give

$$W(T + \delta, a + w) = \sum_{p,q \ge 0} \frac{(-\delta)^p w^q}{p!q!} W_{2p+q}.$$

Take $\delta = \eta^2$ and $w = \sqrt{2}\eta\lambda$. Division by η^m extends holomorphically to $\eta = 0$, with value

$$\frac{2^{m/2} W_m}{m!}\,\mathrm{He}_m(\lambda), \qquad \mathrm{He}_m(\lambda) = m! \sum_{p=0}^{\lfloor m/2 \rfloor} \frac{(-1)^p \lambda^{m-2p}}{2^p p!(m-2p)!}.$$

The identity $\mathrm{He}_m(\lambda) = (-1)^m e^{\lambda^2/2} (d/d\lambda)^m e^{-\lambda^2/2}$ and m integrations by parts give orthogonality to every lower-degree polynomial for the positive Gaussian measure. If there were fewer than m real sign-changing roots, multiplying their linear factors would give a lower-degree polynomial with a nonzero, constant-sign Gaussian pairing with He_m, a contradiction. Thus the Hermite roots are distinct and real. The holomorphic implicit-function theorem gives $\lambda_k(\eta) = \lambda_k + O(\eta)$, proving (4.1). Rouché's theorem on a fixed small disk around a shows these are precisely its m nearby zeros. For $t = T - h$ take $\eta = i\sqrt{h}$. A nonzero Hermite root gives imaginary displacement $\sqrt{2h}\lambda_k + O(h)$, which is nonzero for small positive h. At least one root is nonzero when $m \ge 2$. □

Proposition 4.2 (A positive threshold has a finite collision)**.** *If $\Lambda > 0$, some real $a \ne 0$ satisfies $H(\Lambda, a) = H_x(\Lambda, a) = 0$.*

Proof. Choose $\Lambda/2 \le t_k < \Lambda$ tending to Λ, and a nonreal zero z_k at each t_k. Strip contraction from time zero gives $|\operatorname{Im} z_k| \le 1$. The uniform localization theorem gives $|\operatorname{Re} z_k| < e^{2C/\Lambda}$. A subsequence therefore converges to a finite z_*. Joint continuity gives $H(\Lambda, z_*) = 0$, and the threshold property makes $z_* = a$ real. If this zero were simple, the holomorphic implicit-function theorem would give a unique nearby zero for times near Λ. Reflection forces that zero to be real at real times, contradicting z_k. Hence $H_x(\Lambda, a) = 0$. Equation (2.8) excludes $a = 0$. □

Theorem 4.3 (Exact positive-time endpoint)**.** *For the genuine theta family,*

$$\mathrm{RH} \iff \{(t,x) \in (0,\infty) \times \mathbb{R} : H(t,x) = H_x(t,x) = 0\} = \varnothing. \tag{4.2}$$

It suffices to exclude these common zeros for $0 < t \le 1/2$.

Proof. Under RH, the threshold theorem gives real-rootedness at every $t \ge 0$. A multiple real zero at $T > 0$ would yield a nonreal zero at a slightly earlier positive time by Lemma 4.1, a contradiction. Conversely failure of RH gives $0 < \Lambda \le 1/2$, and Proposition 4.2 supplies the forbidden zero. Zeros at times strictly beyond $1/2$ cannot be multiple by the same splitting argument and de Bruijn's real-rootedness there. □

The criterion makes no simplicity assertion at time zero. Its converse uses uniform finite-height localization; no unproved assertion that collisions cannot escape to infinity has been substituted for it.

5 Stationary Laguerre observations and bounded loss

For real (t, x) write $h_j = \partial_x^j H$ and

$$\mathscr{L}_j = h_{j+1}^2 - h_j h_{j+2}, \qquad \mathcal{E} = H^2 + H_x^2. \tag{5.1}$$

Each h_j obeys the same heat equation. On a differentiable stationary curve $h_{j+1}(t, \chi(t)) = 0$, the chain rule gives

$$\frac{1}{2}\frac{d}{dt} h_j(t, \chi(t))^2 = -h_j h_{j+2} = \mathscr{L}_j. \tag{5.2}$$

We shall use the two levels

$$H_x = 0 \Longrightarrow \mathscr{L}_0 = -H H_{xx} \ge 0, \qquad H_{xx} = 0 \Longrightarrow \mathscr{L}_1 = -H_x H_{xxx} \ge 0. \tag{5.3}$$

5.1 Necessity under real-rootedness

If a real-time slice $f = H(t, \cdot)$ has only real zeros, evenness, the order bound, and $f(0) > 0$ give its paired Hadamard product

$$f(z) = f(0) \prod_{\gamma > 0} (1 - z^2/\gamma^2), \qquad \sum_{\gamma > 0} \gamma^{-2} < \infty, \tag{5.4}$$

with multiplicities. Pairing the genus-one factors cancels their linear exponentials, and evenness removes the remaining exponential linear factor. Finite products p_N are real-rooted and converge locally uniformly, with every fixed derivative. Rolle's theorem gives real-rootedness of their derivatives. For a real-rooted polynomial p, away from its zeros,

$$\frac{(p')^2 - pp''}{p^2} = \sum_{\rho} \frac{m_\rho}{(x-\rho)^2} \ge 0.$$

Continuity handles its zeros. Applying this to $p_N^{(j)}$ and passing to the limit proves $\mathscr{L}_j(t,x) \geq 0$ for every real x and every fixed j. Under RH this applies to all $t \geq 0$, proving the necessity of (5.3) without assuming simplicity at time zero.

5.2 The predecessor branch at every even multiplicity

Lemma 5.1 (Real stationary predecessor)**.** *Let $W_t = -W_{xx}$ be real analytic near (T,a) and have exact even spatial multiplicity $m = 2r$ there. Put $W_k = \partial_x^k W(T,a)$. For small $h = T - t > 0$ there is a real analytic curve $\chi(h)$ with*

$$\chi(h) = a - \frac{W_{m+1}}{rW_m}h + O(h^2), \qquad W_x(T-h,\chi(h)) = 0, \tag{5.5}$$

$$W(T-h,\chi(h)) = \frac{W_m}{r!}h^r + O(h^{r+1}), \qquad W_{xx}(T-h,\chi(h)) = \frac{W_m}{(r-1)!}h^{r-1} + O(h^r). \tag{5.6}$$

In particular $WW_{xx} > 0$ along the predecessor branch.

Proof. The heat Taylor expansion is

$$W(T-h,a+z) = \sum_{p,q\geq 0} \frac{W_{2p+q}}{p!q!}h^p z^q.$$

Set $z = hy$. Because $W_k = 0$ for $k < m$, the quotient $h^{-r}W_x(T-h,a+hy)$ extends analytically to $h = 0$; a nonzero term requires $2p+q+1 \geq 2r$, hence $p+q \geq r$. Its value at zero is

$$\frac{W_m}{(r-1)!}y + \frac{W_{m+1}}{r!}.$$

Its derivative in y is nonzero. The real analytic implicit-function theorem gives $y(h)$ and $\chi(h) = a + hy(h)$, proving (5.5). Substituting in the convergent Taylor series proves (5.6) and

$$WW_{xx} = \frac{W_m^2}{r!(r-1)!}h^{2r-1} + O(h^{2r}) > 0.$$

□

Theorem 5.2 (Two signs exclude finite collisions)**.** *Suppose (5.3) holds throughout an open set $\Omega \subset (0,\infty)\times\mathbb{R}$. Then the genuine theta family has no common zero of H, H_x in Ω.*

Proof. Any such zero has finite multiplicity $m \geq 2$. If m is even, Lemma 5.1 applied to $W = H$ supplies a branch with $H_x = 0$ and $HH_{xx} > 0$, contradicting the first sign. If $m = 2r+1 \geq 3$, apply the lemma to $W = H_x$, of even multiplicity $2r$. Its branch has $H_{xx} = 0$ and $H_xH_{xxx} > 0$, contradicting the second sign. Openness keeps these predecessor branches inside Ω; take $h < T$ so their times remain positive. □

This argument applies to any real analytic heat solution whose candidate zeros have finite spatial multiplicity. Both levels are needed: $W(t,x) = x^3 - 6(t-T)x$ has $\mathscr{L}_0 = 3x^4 + 36(t-T)^2 \geq 0$, yet a triple collision at $(T,0)$; its second stationary sign fails before T. At a common zero, the products HH_{xx} and H_xH_{xxx} vanish. Checking only the collision point therefore loses the contradiction on its predecessor branch. At a double zero, $\mathscr{L}_1 = H_{xx}^2 > 0$; it is not a point of the second stationary set.

Theorem 5.3 (Locally bounded stationary loss)**.** *On an open positive-time domain Ω, suppose nonnegative functions K_0, K_1 are locally bounded, including at every candidate common zero, and satisfy*

$$H_x = 0 \Longrightarrow \mathscr{L}_0 \geq -K_0\mathcal{E}, \qquad H_{xx} = 0 \Longrightarrow \mathscr{L}_1 \geq -K_1\mathcal{E}. \tag{5.7}$$

Then no common zero occurs in Ω.

Proof. At an even collision $m = 2r$, the predecessor branch has $H_x = 0$, $\mathcal{E} = H^2 > 0$, and

$$\frac{-\mathscr{L}_0}{\mathcal{E}} = \frac{H_{xx}}{H} = \frac{r}{h} + O(1).$$

At an odd collision $m = 2r + 1$, its derivative predecessor branch has $H_{xx} = 0$. Lemma 5.1 gives $H_x = ch^r + O(h^{r+1})$ and $H_{xxx} = rch^{r-1} + O(h^r)$, $c \neq 0$. The Taylor series for H at $z = \chi(h) - a = O(h)$ has no term of degree less than $r + 1$ in h, because $2p + q \geq 2r + 1$ forces $p + q \geq r + 1$. Hence $H = O(h^{r+1})$ and $\mathcal{E} = H_x^2(1 + O(h^2)) > 0$. Therefore

$$\frac{-\mathscr{L}_1}{\mathcal{E}} = \frac{H_x H_{xxx}}{H^2 + H_x^2} = \frac{r}{h} + O(1).$$

Either divergence contradicts the assumed local boundedness of the corresponding K_j. All divisions are on nonzero predecessor branches, not at the common zero itself. □

Corollary 5.4 (Global stationary criteria)**.** *For the genuine theta family, RH is equivalent to (5.3) for every $t > 0, x \in \mathbb{R}$. It is also equivalent to existence of nonnegative locally bounded K_0, K_1 on $(0, \infty) \times \mathbb{R}$ satisfying (5.7).*

Proof. Necessity follows from the paired polynomial approximation, with $K_0 = K_1 = 0$. Sufficiency combines either local exclusion theorem with Theorem 4.3. □

Limited-domain estimates exclude only collisions in their covered predecessor neighborhoods. No estimate obtained by dividing by the unproved physical norm establishes bounded loss at its possible zeros.

An equivalent single constrained observation uses the norm's time derivative. Directly from (2.7),

$$\partial_t \mathcal{E} = -2(H H_{xx} + H_x H_{xxx}).$$

It equals $2\mathscr{L}_0$ on $H_x = 0$ and $2\mathscr{L}_1$ on $H_{xx} = 0$. Thus the two stationary signs are precisely

$$\partial_t \mathcal{E} \geq 0 \quad \text{where } H_x H_{xx} = 0. \tag{5.8}$$

Likewise (5.7) is equivalent to existence of one nonnegative locally bounded K with $\partial_t \mathcal{E} \geq -2K\mathcal{E}$ on that union; take $K = \max(K_0, K_1)$ in one direction and $K_0 = K_1 = K$ in the other.

6 Whole-plane scalar curvature criteria

These are the classical complex Laguerre criteria [9, Theorems 2.3–2.4]; the following proof uses the specific even, order-at-most-one theta class rather than any unstated restriction on a general entire function. Fix real time t and set $f(z) = H(t, z)$, $M(x, y) = |f(x + iy)|^2$. Componentwise complex differentiation gives

$$M_y = -2\,\mathrm{Im}(f'\bar{f}), \qquad \frac{1}{2}M_{yy} = |f'|^2 - \mathrm{Re}(f''\bar{f}) =: \mathcal{C}_0(t, x + iy). \tag{6.1}$$

On $y = 0$ this equals $\mathscr{L}_0$, but away from that slice it is a real Hermitian observation rather than the complex bilinear $f'^2 - ff''$.

Theorem 6.1 (Vertical scalar criteria)**.** *For a fixed real theta time, the following conditions are equivalent:*

1. *all zeros of $H(t,\cdot)$ are real;*
2. *$M_y(x,y) \geq 0$ for all real x and $y > 0$;*
3. *$M_{yy}(x,y) \geq 0$ for all real x, y;*
4. *$-\operatorname{Im}(f'(z)/f(z)) \geq 0$ for every upper-half-plane point where the quotient is defined.*

In particular each condition at $t = 0$ is equivalent to RH.

Proof. Under real-rootedness, (5.4) supplies finite real-rooted polynomials converging with their derivatives. Their squared moduli on a vertical line are positive constants times products of $((x-\rho)^2+y^2)$. Each is a polynomial in y^2 with nonnegative coefficients, nondecreasing for $y > 0$ and convex for all real y. Local derivative convergence passes both inequalities to M.

Conversely suppose $f(a+ib) = 0$, $b > 0$. Monotonicity and nonnegativity force $M(a,y) = 0$ for $0 < y < b$, contradicting the identity theorem. For the convexity condition, reflection also gives $M(a,-b) = 0$; convexity between the two zero endpoints, together with nonnegativity, forces the same contradiction. Reflection supplies an upper-half-plane zero whenever any nonreal zero exists.

Away from zeros, $M_y/(2M) = -\operatorname{Im}(f'/f)$. A zero $a+ib$ of multiplicity m produces the pole $m/(z-a-ib)$ plus a locally bounded holomorphic term. At $z = a+i(b+\varepsilon)$ its negative imaginary part is $m/\varepsilon + O(1)$, which changes sign for small positive and negative ε, both still inside the upper half-plane. Thus the fourth condition also excludes all nonreal zeros and is equivalent to the second. □

The growth condition was used only for real-rootedness to imply these signs. It matters: e^{z^2} is real and zero-free but its vertical modulus decreases. Convexity or monotonicity excludes nonreal zeros without a growth assumption. For the theta family the exact endpoint is

$$\mathrm{RH} \iff \mathcal{C}_0(0,z) \geq 0 \quad \text{for every } z \in \mathbb{C}. \tag{6.2}$$

Reality and source positivity make the pure imaginary axis positive already; the equivalent condition requires every translated vertical line. Under real-rootedness the logarithmic derivative also satisfies

$$-\operatorname{Im}\frac{f'(x+iy)}{f(x+iy)} = \sum_{\rho\in\mathbb{R}} \frac{m_\rho y}{(x-\rho)^2+y^2}, \qquad y > 0.$$

This partial-fraction representation is a consequence of the real-zero assumption, not an unconditional arithmetic representation for proving it.

For the trace $f_{\mathrm{jet}} = H + \mathbf{j}H_z$, the scalar observation in (6.2) cannot be replaced by its total norm. Indeed $\frac{1}{2}\partial_y^2|f_{\mathrm{jet}}|^2 = \mathcal{C}_0 + \mathcal{C}_1$, where $\mathcal{C}_1 = |H_{zz}|^2 - \operatorname{Re}(H_{zzz}\overline{H_z})$. The heat solution $H(t,z) = z^2 - 2t + 1$ has nonreal time-zero roots, but its total-norm curvature is $2x^2 + 6y^2 + 2 > 0$. This distinguishes the scalar curvature criterion from an unsigned jet-norm estimate.

7 Hamilton conventions and a regular lift with independent time

We use Hamilton units satisfying

$$\mathbf{i}^2 = \mathbf{j}^2 = \mathbf{k}^2 = -1, \qquad \mathbf{ij} = \mathbf{k}, \quad \mathbf{jk} = \mathbf{i}, \quad \mathbf{ki} = \mathbf{j}.$$

The complex field is the subalgebra $\mathbb{C}_{\mathbf{i}} = \mathbb{R} + \mathbf{i}\mathbb{R}$. Complex coefficients are placed on the right of $\mathbf{j}$. For complex numbers a, b, c, d, direct Hamilton multiplication gives

$$(a + \mathbf{j}b)(c + \mathbf{j}d) = ac - \overline{b}d + \mathbf{j}(bc + \overline{a}d), \qquad \overline{a + \mathbf{j}b} = \overline{a} - \mathbf{j}b. \tag{7.1}$$

Consequently $|a+\mathbf{j}b|^2 = |a|^2+|b|^2$. Throughout this section an overline on a quaternion is quaternion conjugation, and an overline on a complex coefficient is complex conjugation. These operations agree on $\mathbb{C}_{\mathbf{i}}$.

Write the independent quaternion coordinates as

$$q = z_0 + \mathbf{j}z_1, \qquad z_0 = a + \mathbf{i}b, \quad z_1 = c + \mathbf{i}d. \tag{7.2}$$

Since $\mathbf{ji} = -\mathbf{k}$, this is $q = a + \mathbf{i}b + \mathbf{j}c - \mathbf{k}d$. Thus the oriented real Hamilton coordinates are $(q_0, q_1, q_2, q_3) = (a, b, c, -d)$. The normalized left Cauchy–Riemann–Fueter operator and its conjugate are

$$\overline{\partial}_q = \tfrac{1}{2}(\partial_{q_0} + \mathbf{i}\partial_{q_1} + \mathbf{j}\partial_{q_2} + \mathbf{k}\partial_{q_3}) = \tfrac{1}{2}(\partial_a + \mathbf{i}\partial_b + \mathbf{j}\partial_c - \mathbf{k}\partial_d), \tag{7.3}$$

$$\partial_q = \tfrac{1}{2}(\partial_{q_0} - \mathbf{i}\partial_{q_1} - \mathbf{j}\partial_{q_2} - \mathbf{k}\partial_{q_3}). \tag{7.4}$$

All units in these expressions multiply on the left. A field is left Fueter regular if $\overline{\partial}_q F = 0$. This is the usual Hamilton orientation; the minus sign on ∂_d comes solely from $q_3 = -d$. The real definition and classical integral theory are reviewed in [7, 8].

For complex-valued differentiable functions U, V, the coordinate calculation is particularly useful. The a, b terms in $2\overline{\partial}_q(U + \mathbf{j}V)$ equal $2U_{\bar{z}_0} + 2\mathbf{j}V_{z_0}$, and the c, d terms equal $-2V_{z_1} + 2\mathbf{j}U_{\bar{z}_1}$. Hence

$$\overline{\partial}_q(U + \mathbf{j}V) = U_{\bar{z}_0} - V_{z_1} + \mathbf{j}(V_{z_0} + U_{\bar{z}_1}). \tag{7.5}$$

Here subscripts denote ordinary Wirtinger derivatives of the complex coefficients. When applied to a quaternion field, complex coordinate derivatives always act componentwise: $(U+\mathbf{j}V)_{z_0} = U_{z_0}+\mathbf{j}V_{z_0}$. They are not defined by left multiplication of the whole field by $\mathbf{i}$ in a Wirtinger operator. This derivation fixes the convention without importing complex coordinate formulas from a different placement of coefficients.

Let $H(\tau, z)$ be the jointly entire theta heat family defined earlier, with $H_\tau = -H_{zz}$. The real Newman time t remains an independent external parameter. Define

$$\tau = t + \overline{z_1}, \qquad F(t, q) = H(\tau, z_0) + \mathbf{j}H_z(\tau, z_0). \tag{7.6}$$

Both quaternion coordinates are independent. The second coordinate enters through a complex heat-time translation; it does not evaluate the derivative at an independent spectral point.

Proposition 7.1. *For every fixed real t, the field $F(t, \cdot)$ is globally defined, real analytic, and left Fueter regular. It obeys*

$$U_{z_0} = V, \qquad F_t = F_{\bar{z}_1} = -F_{z_0 z_0}, \qquad U = H(\tau, z_0),\ V = H_z(\tau, z_0).$$

Proof. Joint entireness of H makes the composition in (7.6) real analytic on all quaternion space. Holomorphy gives $U_{\bar{z}_0} = V_{z_1} = 0$, while $V_{z_0} = H_{zz}$ and $U_{\bar{z}_1} = H_\tau$. Equation (7.5) therefore becomes $\overline{\partial}_q F = \mathbf{j}(H_{zz} + H_\tau) = 0$. The remaining identities follow by differentiating the declared lift. □

The extension is part of the construction, not a general uniqueness claim for Fueter fields with a given trace. For instance, adding an entire scalar $g(z_1)$, with $g(0) = 0$, preserves regularity, the physical trace, the jet relation, and the external heat equation. It generally changes the declared antiholomorphic dependence on z_1, so later identities requiring that dependence apply to (7.6), not to every such extension. In contrast, the separated expression $H(t, z_0) + \mathbf{j}H_z(t, z_1)$ has regularity defect $-H_{zz}(t, z_1)$; at real $q = x$, evenness gives $H_z(t, 0) = 0$, so that expression also loses the physical first jet.

The operators satisfy

$$\partial_q + \overline{\partial}_q = \partial_{q_0}, \qquad 4\partial_q\overline{\partial}_q = 4\overline{\partial}_q\partial_q = \Delta_4. \tag{7.7}$$

The anticommutation of distinct Hamilton units cancels mixed second derivatives in the product, and each diagonal imaginary term contributes $+\partial^2_{q_\nu}$. On a regular field, $\partial_q F = \partial_{q_0} F$. Every real coordinate derivative preserves regularity, so induction gives

$$G_m := \partial_q^m F = \partial_{q_0}^m F = h_m + \mathbf{j}h_{m+1}, \qquad h_m = \partial_z^m H(\tau, z_0), \qquad \overline{\partial}_q G_m = 0. \tag{7.8}$$

Holomorphy in z_0 identifies the q_0 derivative with the componentwise spectral derivative. The external evolution is therefore

$$\boxed{\overline{\partial}_q F = 0, \qquad F_t = -\partial_q^2 F.} \tag{7.9}$$

These equations retain the declared theta trace and jet constraint; regularity by itself does not impose either of them.

For comparison, on the physical complex pair $u = H(t, z)$, $v = H_z(t, z)$, the jet and heat equations can also be packaged by the matrix differential operator

$$\mathcal{D} = \begin{pmatrix} \partial_z & -1 \\ -\partial_t & -\partial_z \end{pmatrix}, \qquad \mathcal{D}\begin{pmatrix} u \\ v \end{pmatrix} = \begin{pmatrix} u_z - v \\ -u_t - v_z \end{pmatrix} = 0, \qquad \mathcal{D}^2 = (\partial_t + \partial_z^2)I. \tag{7.10}$$

The off-diagonal entries of the square vanish because constant coefficient derivatives commute: the upper-right entry is $-\partial_z + \partial_z = 0$, and the lower-left entry is $-\partial_t\partial_z + \partial_z\partial_t = 0$. Both diagonal entries are $\partial_t + \partial_z^2$. This is an operator on a two-dimensional complex vector, with matrix endomorphisms rather than fixed Hamilton units. It is distinct from the Euclidean Fueter factorization in (7.7); neither factorization alone asserts a selfadjoint realization or a positive physical observation.

8 Physical trace, RH translation, and signed observations

At a real quaternion argument $q = x$, one has $z_0 = x, z_1 = 0$. Since $H(t, x)$ and its derivatives are real,

$$F(t, x) = H(t, x) + \mathbf{j}H_x(t, x), \qquad \mathcal{E}(t, x) := |F(t, x)|^2 = H(t, x)^2 + H_x(t, x)^2. \tag{8.1}$$

Thus $F(t, x) = 0$ is exactly the simultaneous scalar condition $H(t, x) = H_x(t, x) = 0$, namely a multiple real zero of the scalar heat family. A simple zero of H is not a zero of F.

The scalar positive-time collision theorem for the genuine theta family, proved in the scalar part of this manuscript, consequently translates to

$$\boxed{\text{RH} \quad \Longleftrightarrow \quad F(t, x) \neq 0 \quad \text{for all } t > 0,\ x \in \mathbb{R}.} \tag{8.2}$$

No assertion of nonvanishing throughout quaternion space is made, and the restriction $t > 0$ permits multiple scalar zeros at time zero. The quaternion statement uses the scalar collision theorem rather than deducing that theorem from regularity.

The external parameter has an exact translation covariance:

$$F(t,q) = F(0, q + \mathbf{j}t). \tag{8.3}$$

Indeed $q + \mathbf{j}t = z_0 + \mathbf{j}(z_1 + t)$, and real t commutes with conjugation, giving $\overline{z_1 + t} = \bar{z}_1 + t$. For $\mathcal{F}(q) = F(0,q)$, (8.2) becomes

$$\boxed{\text{RH} \quad \Longleftrightarrow \quad \mathcal{F}(x + \mathbf{j}s) \neq 0 \quad (x \in \mathbb{R},\ s > 0).} \tag{8.4}$$

This is a statement on the positive $\mathbf{j}$ half-plane of one explicit entire Fueter field. In the original definition, t remains an independent parameter; translation covariance merely permits the equivalent fixed-field description. The real coordinate x of that half-plane is spectral position, not Newman time.

For the complex observations define $\mathscr{L}_\ell = h_{\ell+1}^2 - h_\ell h_{\ell+2}$. Applying (7.1) to the regular jets gives

$$\overline{G_\ell} G_{\ell+1} = \overline{h_\ell} h_{\ell+1} + \overline{h_{\ell+1}} h_{\ell+2} - \mathbf{j}\mathscr{L}_\ell. \tag{8.5}$$

The scalar complex products are Hermitian, whereas the $\mathbf{j}$ coefficient is the un-conjugated complex Laguerre polynomial. On the physical real slice all h_ℓ are real and

$$\mathscr{L}_\ell(t,x) = \operatorname{Re}(\mathbf{j}\overline{G_\ell(t,x)} G_{\ell+1}(t,x)) = -[\overline{G_\ell(t,x)} G_{\ell+1}(t,x)]_{\mathbf{j}}. \tag{8.6}$$

Here $[Q]_{\mathbf{j}}$ denotes the real coefficient of $\mathbf{j}$ in the Hamilton expansion of Q. Away from the real physical slice $\mathscr{L}_\ell$ is generally complex, so $\operatorname{Re}(\mathbf{j}\overline{G_\ell} G_{\ell+1})$ recovers only its real part.

The scalar projection of the field is also algebraic:

$$P_0(A + \mathbf{j}B) = A, \qquad P_0 Q = \tfrac{1}{2}(Q - \mathbf{i}Q\mathbf{i}). \tag{8.7}$$

It is a projection onto $\mathbb{C}_{\mathbf{i}}$, not onto the real part alone. On the real physical slice its value is real.

The heat equation gives the exact norm evolution

$$\mathcal{E}_t = -2(h_0 h_2 + h_1 h_3), \qquad (\partial_t + \partial_x^2)\mathcal{E} = 2(h_1^2 + h_2^2). \tag{8.8}$$

The latter follows by differentiating $h_0^2 + h_1^2$ twice in x and using $h_{\ell,t} = -h_{\ell+2}$. In particular,

$$h_1 = 0 \Longrightarrow \mathcal{E}_t = 2\mathscr{L}_0, \qquad h_2 = 0 \Longrightarrow \mathcal{E}_t = 2\mathscr{L}_1. \tag{8.9}$$

Thus the two stationary sign hypotheses are exactly equivalent to

$$\mathcal{E}_t \geq 0 \quad \text{on} \quad \{(t,x) : h_1(t,x) h_2(t,x) = 0\}. \tag{8.10}$$

The same set is $(P_0 \partial_q F)(P_0 \partial_q^2 F) = 0$ on the physical slice. A single nonnegative locally bounded function K gives both bounded-loss hypotheses if $\mathcal{E}_t \geq -2K\mathcal{E}$ on that union. Conversely, from separate coefficients K_0, K_1 one can take $K = \max(K_0, K_1)$, wherever both are locally bounded. These reformulations retain the domain and predecessor coverage needed by the collision-exclusion theorem. The positive source term in (8.8) is a backward-heat identity and supplies no forward minimum principle.

The alternative whole-plane curvature criterion must retain the scalar projection. On the complex slice $z_1 = 0$, put $z = x + \mathbf{i}y$. Then $P_0 F(t,z) = H(t,z)$, whereas $|F(t,z)|^2 = |H(t,z)|^2 + |H_z(t,z)|^2$. Define

$$\mathcal{C}_\ell(t,z) = |h_{\ell+1}(t,z)|^2 - \operatorname{Re}(h_{\ell+2}(t,z)\overline{h_\ell(t,z)}).$$

Since $\partial_y h_\ell = \mathbf{i} h_{\ell+1}$ and $\partial_y^2 h_\ell = -h_{\ell+2}$, differentiating the scalar squared modulus gives

$$\mathcal{C}_0(t,z) = \tfrac{1}{2}\partial_y^2 |P_0 F(t,z)|^2, \qquad \tfrac{1}{2}\partial_y^2 |F(t,z)|^2 = \mathcal{C}_0(t,z) + \mathcal{C}_1(t,z). \tag{8.11}$$

Consequently the scalar whole-plane criterion for this theta growth class translates to

$$\mathrm{RH} \quad \Longleftrightarrow \quad \partial_y^2 |P_0 F(0, x + \mathbf{i}y)|^2 \geq 0 \quad \text{for all } x, y \in \mathbb{R}. \tag{8.12}$$

The classical complex Laguerre criterion is stated in [9, Theorem 2.4]; the growth and scalar RH identifications are supplied in the scalar part of this manuscript. Replacing the projection by the total quaternion norm would change the assertion. Indeed $P(\tau, z) = z^2 - 2\tau + 1$ has at $t = 0$

$$\mathcal{C}_0 = 2x^2 + 6y^2 - 2, \qquad \mathcal{C}_1 = 4, \qquad \tfrac{1}{2}\partial_y^2 |F_P(0,z)|^2 = 2x^2 + 6y^2 + 2 > 0,$$

although $P(0, z)$ has the nonreal zeros $\pm\mathbf{i}$.

9 Cauchy–Fueter reconstruction, residuals, and boundary moments

Fix real t. Let Ω be a bounded domain in $\mathbb{H}$, with piecewise C^1 boundary, and let $n(p)$ be its outward unit normal expressed in the oriented Hamilton basis. Set

$$\mathcal{K}(r) = \frac{\bar{r}}{2\pi^2 |r|^4} \qquad (r \neq 0), \qquad D = 2\overline{\partial}_q. \tag{9.1}$$

For smooth quaternion fields A, B, the real divergence theorem, with the multiplication order retained, is

$$\int_{\partial\Omega} B(p) n(p) A(p)\, dS_p = \int_\Omega \big((BD)A + B(DA)\big)\, dV_p, \tag{9.2}$$

where $BD = \sum_{\mu=0}^{3} (\partial_{q_\mu} B) e_\mu$ is the right operator, with $(e_0, e_1, e_2, e_3) = (1, \mathbf{i}, \mathbf{j}, \mathbf{k})$. This is Proposition 7 and Lemma 6 of [8] in our normal notation.

The kernel $\mathcal{K}$ is both left and right regular away from zero and has unit outward flux on every centered sphere:

$$\int_{|r|=\varepsilon} \mathcal{K}(r) \frac{r}{\varepsilon}\, dS_r = 1.$$

Indeed the integrand is the scalar $(2\pi^2\varepsilon^3)^{-1}$ and the sphere has area $2\pi^2\varepsilon^3$. It is the fundamental kernel for the unnormalized operator D; the fundamental kernel for $\overline{\partial}$ is $2\mathcal{K}$. The boundary formula still uses $\mathcal{K}$ with the normalization displayed here.

Apply (9.2) on $\Omega \setminus \overline{B_\varepsilon(q)}$, with $B(p) = \mathcal{K}(p - q)$. The inward orientation on the removed sphere contributes $-A(q)$ in the limit, because A is continuous and the flux is one. Thus every C^1 field A defined near $\overline{\Omega}$ satisfies

$$A(q) = \int_{\partial\Omega} \mathcal{K}(p-q) n(p) A(p)\, dS_p - 2\int_\Omega \mathcal{K}(p-q) \overline{\partial}_p A(p)\, dV_p. \tag{9.3}$$

The volume kernel behaves as $|p-q|^{-3}$ and is locally integrable in four dimensions. This argument asserts no differentiation of that singular volume integral. For regular F, the volume term vanishes and gives the left Cauchy–Fueter formula

$$F(t,q) = \int_{\partial\Omega} \mathcal{K}(p-q) n(p) F(t,p)\, dS_p. \tag{9.4}$$

Its factor order is kernel, normal, field, as in Theorem 9 of [8].

For q in a compact interior set, define

$$\mathcal{K}_m(p;q) = (\partial_{q_0}^m \mathcal{K}(p-q))n(p), \qquad A_m(t,q) = \int_{\partial\Omega} \mathcal{K}_m(p;q)F(t,p)\,dS_p.$$

The boundary kernel is smooth and uniformly separated from its pole, so differentiating (9.4) gives $A_m = G_m$. For $q = x \in \mathbb{R}$, (8.6) becomes the exact signed boundary correlation

$$\begin{aligned}\mathscr{L}_\ell(t,x) &= \operatorname{Re}(\mathbf{j}\overline{A_\ell(t,x)}A_{\ell+1}(t,x)) \\ &= \operatorname{Re}\left[\mathbf{j}\iint_{\partial\Omega\times\partial\Omega} \overline{F(t,p)}\,\overline{\mathcal{K}_\ell(p;x)}\mathcal{K}_{\ell+1}(r;x)F(t,r)\,dS_p\,dS_r\right].\end{aligned} \tag{9.5}$$

Quaternion conjugation reverses the order of the first kernel-field product. This is a signed quadratic observation, not a positive square.

To make the first three observations explicit, take a ball of radius R centered at a real x, and write

$$B(\omega) = F(t, x+R\omega), \qquad \langle B\rangle = \frac{1}{2\pi^2}\int_{S^3} B(\omega)\,dS_\omega.$$

Regularity implies harmonicity by (7.7). The componentwise harmonic Poisson formula is

$$F(t,x+Ry) = \left\langle \frac{1-|y|^2}{|\omega-y|^4}B(\omega)\right\rangle, \qquad |y|<1. \tag{9.6}$$

This is the dimension-four formula in Theorem 34 of [8]. At $y=0$, its kernel and its first two y_0 derivatives are respectively 1, $4\omega_0$, and $24\omega_0^2 - 6$. Thus, differentiating with the boundary sphere fixed,

$$\boxed{A_0 = \langle B\rangle, \qquad A_1 = \frac{4}{R}\langle \omega_0 B\rangle, \qquad A_2 = \frac{6}{R^2}\langle (4\omega_0^2-1)B\rangle.} \tag{9.7}$$

The R^{-1} and R^{-2} factors come from $q_0 = x + Ry_0$. These are harmonic moment formulas, not pointwise equalities between differentiated Cauchy and Poisson kernels. For example, differentiating the Cauchy kernel directly at the center gives $R^{-1}\langle(4\omega_0 - \omega)B\rangle$, which agrees with (9.7) because $\langle \omega B\rangle = 0$ by the regular Stokes theorem.

Stationarity of H or H_x is respectively $P_0A_1 = 0$ or $P_0A_2 = 0$. The physical energy is $|A_0|^2$, and the two Laguerre observations follow from (9.5). For $M = \sup_{\partial B_R}|F|$, the moment formulas give

$$|A_1| \le 4M/R, \qquad |A_2| \le 18M/R^2.$$

The constants use $|\omega_0| \le 1$ and $|6(4\omega_0^2-1)| \le 18$. These upper estimates do not bound the signed correlations from below through a cancellation of A_0. Moreover, the boundary trace samples complex heat times $\tau = t + \bar{z}_1$, so physical real-time signs are not automatically signs of the boundary data.

10 Currents, harmonic norms, orientation, and reciprocal controls

For each regular jet G_ℓ, define the pure-imaginary-quaternion current

$$J_\mu^{(\ell)} = \operatorname{Im}_{\mathbb{H}}(\overline{G_\ell}\,\partial_{q_\mu}G_\ell), \qquad \operatorname{Im}_{\mathbb{H}} Q = Q - \operatorname{Re} Q.$$

Since $\Delta_4 G_\ell = 0$, the product rule gives

$$\sum_{\mu=0}^{3} \partial_{q_\mu} J_\mu^{(\ell)} = \mathrm{Im}_{\mathbb{H}} \left(\sum_{\mu} |\partial_{q_\mu} G_\ell|^2 + \overline{G_\ell} \Delta_4 G_\ell \right) = 0. \tag{10.1}$$

At real physical points, $[J_0^{(\ell)}]_{\mathbf{j}} = -\mathscr{L}_\ell$ by (8.5). Thus a Laguerre sign is an oriented component of a conserved current. Zero closed-boundary flux does not determine its sign at a point.

For the adopted lift, put $e = q_3 = -d$. Then $z_0 = a + \mathbf{i}b$ and $\tau = t + c + \mathbf{i}e$. Differentiation of complex coefficients, with their placement retained, gives

$$F_a = G_1, \quad F_b = G_1\mathbf{i}, \qquad F_c = -G_2, \quad F_e = -G_2\mathbf{i}.$$

For instance $\partial_b(U + \mathbf{j}V) = \mathbf{i}U_{z_0} + \mathbf{j}\mathbf{i}V_{z_0} = (U_{z_0} + \mathbf{j}V_{z_0})\mathbf{i}$. Consequently harmonicity yields

$$\Delta_4 |F|^2 = 2 \sum_{\mu} |\partial_{q_\mu} F|^2 = 4(|h_1|^2 + 2|h_2|^2 + |h_3|^2). \tag{10.2}$$

There is a further logarithmic identity special to the holomorphic pair $(u, v) = (H, H_z)$ in (z_0, τ). Wherever $E = |u|^2 + |v|^2 > 0$, direct complex differentiation gives, for $\alpha = z_0$ or τ,

$$\partial_\alpha \partial_{\bar{\alpha}} \log E = \frac{E(|u_\alpha|^2 + |v_\alpha|^2) - |\bar{u}u_\alpha + \bar{v}v_\alpha|^2}{E^2} = \frac{|uv_\alpha - vu_\alpha|^2}{E^2}.$$

The two determinants are $uv_{z_0} - vu_{z_0} = -\mathscr{L}_0$ and $uv_\tau - vu_\tau = h_1 h_2 - h_0 h_3 = \partial_z \mathscr{L}_0$. Since $\Delta_4 = 4(\partial_{z_0}\partial_{\bar{z}_0} + \partial_\tau \partial_{\bar{\tau}})$ at fixed external time,

$$\boxed{\Delta_4 \log |F|^2 = \frac{4(|\mathscr{L}_0|^2 + |\partial_z \mathscr{L}_0|^2)}{|F|^4}.} \tag{10.3}$$

This statement is restricted to $F \neq 0$ and uses the declared two-variable holomorphic pair, beyond general Fueter regularity. Both norm identities erase the orientation by taking squares; subharmonicity of a norm or logarithmic norm does not exclude zeros.

The real Jacobian has a fixed orientation sign. With domain coordinates (a, b, c, e), the pair $(z_0, \tau) = (a + \mathbf{i}b, t + c + \mathbf{i}e)$ has the usual complex orientation. Its complex derivative matrix is

$$\begin{pmatrix} h_1 & -h_2 \\ h_2 & -h_3 \end{pmatrix}, \qquad \det_{\mathbb{C}} = h_2^2 - h_1 h_3 = \mathscr{L}_1.$$

The real determinant of a complex-linear map is the squared modulus of its complex determinant. But the quaternion output coordinates of $u + \mathbf{j}v$ are $(\mathrm{Re}\, u, \mathrm{Im}\, u, \mathrm{Re}\, v, -\mathrm{Im}\, v)$, reversing one coordinate. Therefore

$$\det_{\mathbb{R}}(d_q F) = -|\mathscr{L}_1(\tau, z_0)|^2. \tag{10.4}$$

The complex determinant in the opposite variable order (τ, z_0) is $-\mathscr{L}_1$. At a double physical zero, $h_0 = h_1 = 0$ and $h_2 \neq 0$, so $\mathscr{L}_1 = h_2^2 \neq 0$. Such a scalar collision is a nondegenerate zero of the quaternion map, not a forbidden degeneracy.

Conjugating the left equation gives

$$\overline{F}\, D^- := \partial_{q_0}\overline{F} - \partial_{q_1}\overline{F}\,\mathbf{i} - \partial_{q_2}\overline{F}\,\mathbf{j} - \partial_{q_3}\overline{F}\,\mathbf{k} = 0.$$

Thus $\overline{F}$ is right regular for the conjugate operator; it is not generally right regular for D. In (9.2), substituting $B = \overline{F}, A = F$ leaves the volume term $(\overline{F}D)F$. Dropping it would not be justified.

Pointwise reciprocal and logarithmic derivative arguments likewise require care. The entire backward-heat solution $H(\tau, z) = z$ has the regular, nowhere-zero lift $F = z_0 + \mathbf{j}$, but

$$F^{-1} = \frac{\bar{z}_0 - \mathbf{j}}{1 + |z_0|^2}, \qquad \overline{\partial}_q(F^{-1}) = \frac{1 + \mathbf{j}\bar{z}_0}{(1 + |z_0|^2)^2} \neq 0. \tag{10.5}$$

For this calculation, write $U = \bar{z}_0/(1 + |z_0|^2)$, $V = -1/(1 + |z_0|^2)$ in (7.5); then $U_{\bar{z}_0} = (1 + |z_0|^2)^{-2}$ and $V_{z_0} = \bar{z}_0(1 + |z_0|^2)^{-2}$. Here $F^{-1}\partial_q F = F^{-1}$ is also nonregular. The general failure of ordinary products and compositions to preserve Fueter regularity is discussed in [7]; the explicit control supplies the reciprocal obstruction needed here.

On the real physical slice only, whenever $F \neq 0$,

$$F^{-1}F_x = \tfrac{1}{2}\partial_x \log \mathcal{E} - \mathbf{j}\frac{\mathscr{L}_0}{\mathcal{E}}. \tag{10.6}$$

This is an algebraic identity obtained by dividing (8.5) by $\mathcal{E}$; it asserts no ambient regularity of the quotient. Locally writing $F = \sqrt{\mathcal{E}}\, e^{\mathbf{j}\theta}$ identifies its $\mathbf{j}$ component with the phase derivative $\theta_x = -\mathscr{L}_0/\mathcal{E}$.

Finally, the backward-heat polynomial $P(\tau, z) = z^2 - 2\tau + 1$ gives

$$F_P = z_0^2 - 2t - 2\bar{z}_1 + 1 + 2\mathbf{j}z_0, \qquad A_0 = x^2 - 2t + 1 + 2\mathbf{j}x, \quad A_1 = 2x + 2\mathbf{j}, \quad A_2 = 2.$$

Equations (8.6) and (10.4) give $\mathscr{L}_0 = 2x^2 + 4t - 2$, $\mathscr{L}_1 = 4$, and $\det d_q F_P = -16$. The physical field nevertheless vanishes at $(t, x) = (1/2, 0)$. This control satisfies the lift, regularity, heat, and boundary identities, but it violates the first stationary sign before its collision. The equations therefore organize the RH observation without replacing the needed estimate from the complete theta source.

11 A complete theta estimate on a bounded spectral domain

For clarity, all source variables in this section are denoted by u. The physical spectral coordinate is x, and t is independent real Newman time. We use the normalization

$$\phi_n(u) = (2\pi^2 n^4 e^{9u} - 3\pi n^2 e^{5u}) e^{-\pi n^2 e^{4u}}, \qquad u \geq 0, \quad n \geq 1, \tag{11.1}$$

$$\Phi(u) = \sum_{n \geq 1} \phi_n(u), \qquad H(t, x) = \int_0^\infty e^{tu^2} \Phi(u) \cos(xu)\, du. \tag{11.2}$$

Thus $H(0, z) = \xi(1/2 + iz/2)/8$. The complete source extends to an even smooth function on the real line. In particular, every odd endpoint derivative of Φ vanishes. One direct way to see this is to put

$$h(u) = e^u \Theta(e^{4u}, 0), \qquad \Theta(r, y) = \sum_{n \in \mathbb{Z}} e^{-\pi n^2 r + 2\pi i n y}.$$

Jacobi inversion gives $h(-u) = h(u)$, and differentiation of its normally convergent series gives $\Phi = (h'' - h)/16$ on $u \geq 0$. Every fixed source derivative has superexponential decay at infinity. Consequently the cosine transform in (11.2) is entire in (t, x), differentiation under its integral is legitimate, and its real spectral restriction is Schwartz for each fixed real t. No finite theta truncation is made here.

For $0 \leq t \leq 1$, define

$$w_t(u) = e^{tu^2} \Phi(u), \qquad M_j(t) = \int_0^\infty u^j w_t(u)\, du, \qquad \delta = \frac{1}{120000}.$$

Every $\phi_n(u)$ is strictly positive on this half-line: writing $q = \pi n^2 e^{4u}$, its factor $2q - 3$ is positive.

Theorem 11.1 (Complete-source central moment estimate)**.** *For every* $0 \le t \le 1$*,*

$$M_2(t) - \frac{25}{2} M_4(t) > \frac{2413193}{268912000000} > \delta. \tag{11.3}$$

Both the entire theta-index sum and the negative integration tail are included in this estimate.

Proof. First consider $0 \le u \le 1/10$. With $q = \pi e^{4u}$,

$$\phi_1(u) = e^u g(q), \qquad g(q) = q(2q-3)e^{-q}, \qquad g'(q) = -(2q-1)(q-3)e^{-q}.$$

The elementary bounds $\pi < 22/7$ and $e^{2/5} < 3/2$ give $3 < q < 33/7$. Hence g is decreasing on the required interval, and $e < 11/4$, $e^{2/7} > 9/7$ give

$$\Phi(u) \ge \phi_1(u) > g(33/7) > \frac{1485}{49}\frac{9}{7}\left(\frac{4}{11}\right)^5 > \frac{6}{25}. \tag{11.4}$$

Only positive summands have been discarded.

For the complete upper tail let $y = \pi e^{4u} \ge \pi$. The ratio between successive terms of $n^4 e^{-yn^2}$ is at most

$$16e^{-3y} \le 16e^{-3\pi} < \frac{1}{2}.$$

Dropping the negative part of each summand in (11.1) and summing its full geometric majorant yields

$$\Phi(u) < 4\pi^2 e^{9u - \pi e^{4u}}, \qquad u \ge 0. \tag{11.5}$$

Set $b = 7/25$ and $L_t(u) = 9u + tu^2 - \pi e^{4u}$. Using $\pi > 25/8$ and $e^{28/25} > 153/50$, one obtains

$$L_t(b) < \frac{1624}{625} - \frac{153}{16} < -\frac{69}{10}, \qquad L_t'(b) < 9 + \frac{14}{25} - \frac{153}{4} < -28.$$

Moreover $L_t''(u) = 2t - 16\pi e^{4u} < 0$ for $u \ge b$. Thus $L_t(u) < -69/10 - 28(u-b)$ on the full tail. Finally $4\pi^2 < 40$ and

$$e^{69/10} > \left(\frac{8}{3}\right)^6 \left(1 + \frac{9}{10} + \frac{1}{2}\left(\frac{9}{10}\right)^2\right) > 800$$

prove the uniform envelope

$$w_t(u) < \frac{1}{20} e^{-28(u-b)}, \qquad u \ge b. \tag{11.6}$$

On $[0, b]$, the factor $u^2(1 - 25u^2/2)$ is nonnegative. On $u \ge b$ it is bounded below by $-25u^2(u^2 - b^2)/2$, because $1 - 25b^2/2 = 1/50$. Apply (11.4) to the positive core, discard the nonnegative remainder in $[0, b]$, and pay the whole negative tail using (11.6). This gives

$$M_2 - \frac{25}{2} M_4 > \frac{6}{25} \int_0^{1/10} u^2(1 - 25u^2/2)\, du - \frac{5}{8} I,$$

where, after the substitution $u = b + s$,

$$\begin{aligned} I &= \int_b^\infty u^2(u^2 - b^2) e^{-28(u-b)}\, du \\ &= \frac{2b^3}{28^2} + \frac{10b^2}{28^3} + \frac{24b}{28^4} + \frac{24}{28^5} = \frac{3497259}{33614000000}. \end{aligned}$$

The retained difference is exactly

$$\frac{37}{500000} - \frac{5}{8} I = \frac{2413193}{268912000000} > \frac{1}{120000}.$$

This proves (11.3). □

Corollary 11.2 (Physical norm and stationary signs on the central domain)**.** *For* $0 \le t \le 1$,

$$H_{xx}(t,x) < -\delta \qquad (|x| \le 5), \tag{11.7}$$

$$-H_x(t,x) > x\delta \qquad (0 < x \le 8), \tag{11.8}$$

$$H(t,x) > \frac{7733}{350000} > \frac{1}{50} \qquad (|x| \le 1), \tag{11.9}$$

$$|H(t,x) + \mathbf{j}H_x(t,x)| > \delta \qquad (|x| \le 8). \tag{11.10}$$

On $|x| \le 5$*, the only first stationary point is* $x = 0$*, and the second stationary set is empty. At that first stationary point,*

$$\mathscr{L}_0(t,0) = H(t,0)M_2(t) > \frac{1}{5000000}. \tag{11.11}$$

Proof. For every real y, $\cos y \ge 1 - y^2/2$. Integrating this inequality from zero gives $\sin y \ge y - y^3/6$ for every $y \ge 0$. Because w_t is positive, these global inequalities imply

$$-H_{xx} \ge M_2 - \frac{x^2}{2}M_4, \qquad -H_x \ge x\left(M_2 - \frac{x^2}{6}M_4\right) \quad (x > 0).$$

The first right side exceeds δ for $|x| \le 5$. The second exceeds $x\delta$ for $0 < x \le 8$, since $8^2/6 < 25/2$. Evenness handles negative x. These estimates keep the oscillatory source tail rather than replacing its cosine or sine by a presumed sign.

For $|x| \le 1$, the core satisfies $\cos(xu) \ge 199/200$ on $0 \le u \le 1/10$. The interval $1/10 \le u \le b$ has positive cosine. The remaining tail costs at most $1/560$, by (11.6). Therefore

$$H(t,x) > \frac{6}{25}\frac{1}{10}\frac{199}{200} - \frac{1}{560} = \frac{7733}{350000} > \frac{1}{50}.$$

Use this scalar component for $|x| \le 1$, and (11.8) for $1 \le |x| \le 8$, to prove (11.10). Strict concavity and evenness give the assertion about stationary sets. Finally $H(t,0) > 3/125$ follows by integrating the positive core alone, and $M_2 > \delta$ follows from (11.3). Their product is greater than $1/5000000$. □

The bounded range $|x| \le 8$ is part of the theorem. It does not exclude collisions at unknown spectral heights.

11.1 The second stationary sign throughout the central domain

Proposition 11.3 (A fourth–sixth moment gap)**.** *For* $0 \le t \le 1$,

$$M_4 - \frac{32}{3}M_6 > \frac{2092283587}{3853516800000000} > \frac{1}{2000000}. \tag{11.12}$$

Consequently, for $0 < |x| \le 8$,

$$\mathscr{L}_1(t,x) > \frac{x^2}{240000000000}. \tag{11.13}$$

On $|x| \le 8$*, the first stationary set is exactly* $x = 0$*, with the margin in* (11.11)*; the second stationary sign holds wherever* $H_{xx} = 0$.

Proof. The decreasing function $g(q) = q(2q-3)e^{-q}$ used above gives two additional source cores:

$$\Phi(u) > 3/25 \quad (0 \le u \le 3/20), \qquad \Phi(u) > 1/20 \quad (0 \le u \le 1/5).$$

Indeed $e^{3/5} < 11/6$, $\pi < 22/7$ imply $q < 6$ on the first interval, and $g(6) > 54(4/11)^6 > 3/25$. On the second, $e^{4/5} < 9/4$ gives $q < 36/5$; using $e < 11/4$, $e^{1/5} < 5/4$,

$$g(36/5) > \frac{2052}{25}(4/11)^7(4/5) > 1/20.$$

Keep $\Phi > 6/25$ on the first interval $[0, 1/10]$.

For the full upper tail set $b = 3/10$. The rational bounds $\pi > 333/106$, $e^{6/5} > 83/25$ give, uniformly for $t \le 1$,

$$L_t(b) < -40491/5300 < -38/5, \qquad L_t'(b) < -42558/1325 < -32.$$

Since $L_t'' < 0$, $4\pi^2 < 40$, and $e^{38/5} > 1960$, the complete-index envelope is

$$w_t(u) < \frac{1}{49}e^{-32(u-b)} \qquad (u \ge b).$$

Put $f(u) = u^4(1 - 32u^2/3)$. This is nonnegative on $[0, b]$, and on the entire tail it is bounded below by $-(32/3)u^4(u^2 - b^2)$. Hence

$$M_4 - \frac{32}{3}M_6 > P - \frac{32}{147}I,$$

where the three successive positive cores and the full negative tail give

$$P = \frac{6}{25}\int_0^{1/10} f + \frac{3}{25}\int_{1/10}^{3/20} f + \frac{1}{20}\int_{3/20}^{1/5} f = \frac{111799}{33600000000},$$
$$I = \int_b^\infty u^4(u^2 - b^2)e^{-32(u-b)}\,du = \frac{85837761}{6710886400000}.$$

Their exact difference proves (11.12). The exponential comparisons follow from finite rational Taylor sums with geometric remainder bounds; the sharper bound on π follows from the Machin sums described in the appendix.

For $0 < x \le 8$, the global sine inequality and source positivity give

$$h_3 \ge x(M_4 - x^2M_6/6) > x/2000000.$$

Combine this with $-h_1 > x/120000$ to obtain $h_2^2 - h_1h_3 > x^2/240000000000$. Evenness transfers the sign to negative heights. The central slope excludes every nonzero first stationary point. At zero, $h_2 = -M_2 < 0$, so the second stationary set does not meet the origin. □

12 Full angular gluing and absolute observation stability

Let $(c_n)_{n\in\mathbb{Z}}$ be a bounded real sequence satisfying $c_{-n} = c_n$ and $c_0 = 1$. Its prepared theta field is

$$\Theta_c(r, y) = \sum_{n\in\mathbb{Z}} c_n e^{-\pi n^2 r + 2\pi i n y}, \qquad r > 0.$$

Boundedness guarantees normal convergence on compact angular sets. The complete imaginary elliptic defect is

$$E_c(r, y) = e^{-\pi r}\left[\Theta_c(r, y + ir) - e^{\pi r - 2\pi i y}\Theta_c(r, y)\right]. \tag{12.1}$$

We use its full angular norm at the *single* scale $r = 1$. This is an infinite Fourier observation, not a finite jet condition or an estimate over all theta scales.

Proposition 12.1 (Exact defect norm and balanced gauge). *Put $d_m = c_m - c_{m-1}$ for $m \geq 1$ and $w_m = e^{-2\pi m^2} + e^{-2\pi(m-1)^2}$. Then*

$$D^2 := \int_0^1 |E_c(1,y)|^2\, dy = \sum_{m\geq 1} |d_m|^2 w_m. \tag{12.2}$$

For $\mathcal{B}_c(0,z) = e^{\pi z^2/2}\Theta_c(1,z)$,

$$D^2 = \int_0^1 e^{-\pi(1+y^2)} \left|\mathcal{B}_c(0,y+i) - e^{\pi/2-\pi iy}\mathcal{B}_c(0,y)\right|^2 dy. \tag{12.3}$$

In particular $D = 0$ forces $c_n = 1$ for every n.

Proof. Termwise substitution into (12.1), followed by the index change in the second term, gives the normally convergent series

$$E_c(r,y) = \sum_{n\in\mathbb{Z}} (c_n - c_{n+1}) e^{-\pi(n+1)^2 r} e^{2\pi i n y}.$$

Parseval on $[0,1]$ gives its squared coefficient norm. For $n = m-1 \geq 0$, the coefficient difference is $-d_m$ and its weight is $e^{-2\pi m^2}$. For $n = -m \leq -1$, symmetry gives the difference d_m and the weight $e^{-2\pi(m-1)^2}$. Their sum proves (12.2). Every weight is positive; $c_0 = 1$ proves the rigidity statement.

For real y, direct multiplication of the balanced gauge gives

$$E_c(1,y) = e^{-\pi-\pi(y+i)^2/2}\left[\mathcal{B}_c(0,y+i) - e^{\pi/2-\pi iy}\mathcal{B}_c(0,y)\right].$$

The squared modulus of the prefactor is $e^{-\pi(1+y^2)}$, proving (12.3). □

The stability observation below is explicitly defined on the half-line:

$$\Phi_c^+(u) = \sum_{n\geq 1} c_n \phi_n(u), \qquad H_c(t,x) = \int_0^\infty e^{tu^2}\Phi_c^+(u)\cos(xu)\, du. \tag{12.4}$$

For general c, its even source extension need not be smooth at zero or equal the projection of the full modular prepared field. It is nonetheless entire in (t,x), obeys backward heat, and is real and even in real x. If all $c_n > 0$, its source is positive.

Theorem 12.2 (Four-jet gluing stability). *For every real x, $0 \leq t \leq 1$, and $j = 0,1,2,3$,*

$$|\partial_x^j (H_c - H)(t,x)| \leq D C_j, \qquad C_j^2 = \sum_{m\geq 1} \frac{A_{jm}^2}{w_m}, \tag{12.5}$$

where the full positive sums are

$$A_{jm} = \sum_{n\geq m} \frac{2\pi^2 n^4 j! e^{-\pi n^2}}{(4\pi n^2 - 9)^{j+1}}.$$

These finite constants satisfy

$$C_0 < \frac{1}{2}, \qquad C_1 < \frac{1}{8}, \qquad C_2 < \frac{1}{12}, \qquad C_3 < \frac{1}{12}. \tag{12.6}$$

The same four rational constants bound the genuine jets $|\partial_x^j H|$ on this domain.

Proof. Let $\Delta h_j = \partial_x^j(H_c - H)$. Using $c_n - 1 = \sum_{m=1}^{n} d_m$, the complete series can be regrouped as

$$\Delta h_j = \sum_{m\ge 1} d_m B_{jm}, \qquad B_{jm} = \sum_{n\ge m} \int_0^\infty e^{tu^2}\phi_n(u)u^j \cos(xu + j\pi/2)\,du.$$

Absolute convergence justifies this regrouping: the coefficient sequence is bounded, so $\sum_{m\le n}|d_m| = O(n)$, while the moment bounds below decay as $e^{-\pi n^2}$. For $u \ge 0$, $e^{4u} \ge 1 + 4u + 8u^2$. Drop the negative summand in ϕ_n, and note $t - 8\pi n^2 < 0$:

$$e^{tu^2}\phi_n(u) \le 2\pi^2 n^4 e^{-\pi n^2} e^{-(4\pi n^2 - 9)u}.$$

Integrating u^j against this full half-line envelope proves $|B_{jm}| \le A_{jm}$. Cauchy–Schwarz against the exact weights in (12.2) proves (12.5).

To bound the constants, set

$$a_j = \frac{2\pi^2 j!}{(4\pi - 9)^{j+1}}, \quad \rho_0 = 4e^{-3\pi}, \quad \rho_j = e^{-3\pi}\ (j \ge 1), \quad r_* = e^{-4\pi}.$$

Since $4\pi n^2 - 9 \ge (4\pi - 9)n^2$, the summand in A_{jm} is at most $a_j n^{2-2j} e^{-\pi n^2}$. Successive ratios of this majorant are bounded by ρ_j, giving

$$A_{jm} \le \frac{a_j m^{2-2j} e^{-\pi m^2}}{1 - \rho_j}.$$

Because $w_m \ge e^{-2\pi(m-1)^2}$, this implies

$$C_0^2 \le \left(\frac{a_0 e^{-\pi}}{1 - \rho_0}\right)^2 \frac{1 + 11r_* + 11r_*^2 + r_*^3}{(1 - r_*)^5},$$
$$C_j^2 \le \left(\frac{a_j e^{-\pi}}{1 - \rho_j}\right)^2 \frac{1}{1 - r_*}, \qquad j = 1, 2, 3.$$

Here the first identity uses $\sum_{m\ge 1} m^4 r^{m-1} = (1 + 11r + 11r^2 + r^3)/(1 - r)^5$; for $j \ge 1$, $m^{4-4j} \le 1$. The elementary bounds

$$3 < \pi < 22/7, \quad e^3 > 20, \quad e^{-\pi} < 1/20, \quad \rho_0 < 1/2000, \quad \rho_j < 1/8000, \quad r_* < 1/160000$$

reduce the asserted strict rational bounds to finite rational comparisons, detailed in the reproducibility appendix. Finally the genuine moment bound is $|\partial_x^j H| \le A_{j1}$; the same majorants give its stated rational bounds. □

Corollary 12.3 (Absolute signed-form stability and a robust central gap)**.** *Let* $\mathscr{L}_j(f) = (f^{(j+1)})^2 - f^{(j)} f^{(j+2)}$. *On every real* x *and* $0 \le t \le 1$,

$$|\mathscr{L}_0(H_c) - \mathscr{L}_0(H)| \le \frac{11}{192}(2D + D^2), \tag{12.7}$$
$$|\mathscr{L}_1(H_c) - \mathscr{L}_1(H)| \le \frac{5}{288}(2D + D^2). \tag{12.8}$$

If $D \le 1/30000$, *then*

$$|H_c(t, x) + \mathbf{j}H_{c,x}(t, x)| > \frac{1}{240000}, \qquad |x| \le 8, \quad 0 \le t \le 1. \tag{12.9}$$

On $|x| \le 5$, $H_{c,xx} < -1/180000$; *the only first stationary point is zero, the second stationary set is empty, and* $\mathscr{L}_0(H_c; t, 0) > 1/9000000$.

Proof. Write $S_0 = 1/2, S_1 = 1/8, S_2 = S_3 = 1/12$. Expansion of the quadratic Laguerre form, followed by $|h_j| < S_j$ and $|\Delta h_j| \le DS_j$, bounds its error by

$$(S_{j+1}^2 + S_j S_{j+2})(2D + D^2).$$

The two coefficients are $11/192$ and $5/288$. This comparison is at the same spectral point; it does not assume that stationary sets of H and H_c coincide.

For the norm bound, on $|x| \le 1$ the scalar component satisfies

$$H_c > 1/50 - D/2 > 1/100.$$

On $1 \le |x| \le 8$, the derivative component satisfies

$$|H_{c,x}| > \delta - D/8 \ge 1/240000.$$

The curvature transfer gives $H_{c,xx} < -\delta + D/12 \le -1/180000$. Evenness now identifies the stationary sets. At zero $H_c > 3/125 - D/2 > 1/50$, and multiplication of the last curvature gap by $1/50$ proves the stated Laguerre margin. □

The estimates (12.7)–(12.8) transfer a separately established margin. They do not supply the genuine theta sign, or a loss bounded relative to $H^2 + H_x^2$.

13 The scope of an approximate gluing certificate

Proposition 13.1 (Arbitrarily small defects with remote joint zeros)**.** *Fix $T \in (0,1)$. There are bounded positive symmetric coefficient sequences with $c_0 = 1$, defects $D_X > 0$ tending to zero, and half-line observations (12.4) satisfying*

$$H_c(T, X) = H_{c,x}(T, X) = 0$$

at an unbounded sequence of real heights X. The coefficient changes and D_X decrease faster than every inverse power of X. These observations fail exact imaginary gluing and smooth modular reflection; they are not genuine theta counterexamples.

Proof. Put $K_n(x) = \int_0^\infty e^{Tu^2} \phi_n(u) \cos(xu)\, du$, and $\alpha_n = \pi n^2$. The polynomial recurrence

$$P_0(q) = 2q - 3, \qquad P_{k+1}(q) = (5 - 4q)P_k(q) + 4qP_k'(q)$$

gives $\phi_n^{(k)}(0) = \alpha_n e^{-\alpha_n} P_k(\alpha_n)$. In particular,

$$P_1(q) = -8q^2 + 30q - 15, \quad P_3(q) = -128q^4 + 1440q^3 - 4232q^2 + 3270q - 375.$$

Write $b_n = \phi_n'(0)$ and $e_n = \phi_n'''(0)$. Repeated integration by parts, with all derivatives integrable, gives

$$K_n = -b_n x^{-2} + (e_n + 6Tb_n)x^{-4} + O(x^{-6}),$$
$$K_n' = 2b_n x^{-3} - 4(e_n + 6Tb_n)x^{-5} + O(x^{-7}).$$

The derivative expansion can be obtained independently from its sine integral, so no differentiation of an unspecified remainder is needed. Consequently

$$\det \begin{pmatrix} K_1 & K_2 \\ K_1' & K_2' \end{pmatrix} = 8\pi^2 e^{-5\pi} P(\pi) x^{-7} + O(x^{-9}),$$

where

$$P(q) = -113400q + 907200q^2 - 2570400q^3 + 3081600q^4 - 1520640q^5 + 245760q^6.$$

This polynomial is $P_1(q)P_3(4q) - P_1(4q)P_3(q)$. Exact rational interval evaluation gives $4.598266 < P(\pi) < 4.598268$, as detailed in the appendix. The matrix is therefore invertible for every sufficiently large x, and its inverse grows at most polynomially.

Choose an unbounded sequence of sufficiently large X with $H(T, X) \neq 0$, possible because $H(T, \cdot)$ is nonzero entire. Solve

$$\begin{pmatrix} K_1(X) & K_2(X) \\ K_1'(X) & K_2'(X) \end{pmatrix} \begin{pmatrix} a_X \\ b_X \end{pmatrix} = -\begin{pmatrix} H(T, X) \\ H_x(T, X) \end{pmatrix}.$$

The right side is Schwartz, so a_X, b_X decrease faster than every inverse power of X. Let $c_1 = 1 + a_X, c_2 = 1 + b_X$, retain $c_n = 1$ for $n \geq 3$, and impose symmetry and $c_0 = 1$. For large X all coefficients are positive and the defining system gives the exact joint zero. The coefficient change is nonzero, and

$$D_X^2 = a_X^2 w_1 + (b_X - a_X)^2 w_2 + b_X^2 w_3 > 0$$

tends to zero at the same superalgebraic rate. The unheated odd-jet matrix $\left(\begin{smallmatrix} b_1 & b_2 \\ e_1 & e_2 \end{smallmatrix}\right)$ has determinant $4\pi^2 e^{-5\pi} P(\pi) \neq 0$. Thus the nonzero perturbation cannot leave both the first and third odd endpoint jets zero. Its even extension is not smooth, and cannot be a smooth modular projection. □

14 Signed paired kernels and endpoint compensation

For real t, x, put $g_{n,t}(u) = e^{tu^2}\phi_n(u)$. Direct expansion of the physical Laguerre forms gives the exact complete paired identity

$$\mathscr{L}_j(t, x) = \frac{1}{4} \sum_{n,m \geq 1} \iint_{(0,\infty)^2} g_{n,t}(u) g_{m,t}(v) (uv)^j \times \left[(u+v)^2 \cos x(u-v) + (-1)^j (u-v)^2 \cos x(u+v)\right] du\, dv, \qquad j = 0, 1. \tag{14.1}$$

Absolute convergence follows from the complete theta source moments through order four. To derive it, write $H^{(j)} = \sum_n \int u^j g_{n,t}(u) \cos(xu + j\pi/2)\, du$, symmetrize $H^{(j)} H^{(j+2)}$ in u, v, and use the product-to-sum identities. Both terms, and all off-diagonal indices $n \neq m$, remain. Substitution of the same source into Fueter boundary reproduction produces this identity: the boundary radius disappears under exact reproduction, rather than creating a new positive multiplier.

The individual scalar channel $H_{n,t} = \int g_{n,t} \cos(xu)\, du$ has an algebraic endpoint tail:

$$H_{n,t}(x) = -b_n x^{-2} + O(x^{-4}), \qquad b_n = \phi_n'(0).$$

Repeated endpoint expansions of its derivatives give, when $b_n \neq 0$,

$$\mathscr{L}_j(H_{n,t}; x) = -(j+2)((j+1)!)^2 b_n^2 x^{-2j-6} + O(x^{-2j-8}), \qquad j = 0, 1. \tag{14.2}$$

Indeed the leading coefficient is $(j+2)!^2 - (j+1)!(j+3)! = -(j+2)((j+1)!)^2$. Thus separate diagonal channels are eventually negative. For the complete source, in contrast,

$$\sum_{n \geq 1} \phi_n^{(2k+1)}(0) = 0, \qquad k \geq 0.$$

The full heated source has the same cancellation of every odd endpoint jet, because e^{tu^2} is even. This is necessary compensation for its Schwartz Fourier decay. It rules out a global proof that treats individual diagonal channels as nonnegative or ignores endpoint cancellation in a theta truncation.

Proposition 14.1 (Compensated all-index truncation)**.** *Fix a positive integer K. There exist odd Schwartz functions $\beta_0, \dots, \beta_{K-1}$ on $\mathbb{R}$, each an odd polynomial times e^{-u^2}, satisfying*

$$\beta_k^{(2\ell+1)}(0) = \delta_{k\ell}, \qquad 0 \le k, \ell < K.$$

Define, on $u \ge 0$,

$$\widetilde{g}_{n,t}(u) = g_{n,t}(u) - \sum_{k=0}^{K-1} g_{n,t}^{(2k+1)}(0)\beta_k(u), \qquad R_N(u) = \sum_{n>N} \widetilde{g}_{n,t}(u).$$

Then $\sum_{n\ge1} \widetilde{g}_{n,t} = e^{tu^2}\Phi$, and, for $j = 0, 1, 2, 3$, $x \ne 0$,

$$\left| \int_0^\infty u^j R_N(u) \cos(xu + j\pi/2)\, du \right| \le |x|^{-2K} \left\| (u^j R_N)^{(2K)} \right\|_{L^1(0,\infty)}. \tag{14.3}$$

The norm on the right includes every omitted index and is finite. The following explicit majorant gives an all-index payment for it.

Proof. The jets of $u^{2k+1}e^{-u^2}$ form a triangular matrix, with nonzero diagonal, so finite triangular interpolation constructs the β_k. For example $\beta_0 = ue^{-u^2}$ when $K = 1$. Every compensated summand has its first K odd jets zero. Absolute convergence of the endpoint-jet sums and full-source cancellation show that their complete sum is unchanged. For even j, the cosine integral in (14.3) has boundary terms involving odd derivatives of $u^j R_N$; for odd j, its sine integral has boundary terms involving even derivatives. Each relevant derivative below order $2K$ is zero, because its surviving derivative of R_N is odd and below $2K$. All upper endpoints vanish. Repeating integration by parts $2K$ times therefore proves (14.3). □

Here is an explicit positive-polynomial majorant for the remainder norm; it avoids leaving a finite theta cutoff unpaid. Put $\alpha_n = \pi n^2$, $q = \alpha_n e^{4u}$. The derivative representation

$$g_{n,t}^{(k)}(u) = \alpha_n e^{tu^2+5u-q} P_{k,t}(u, q)$$

has the exact recurrence

$$P_{0,t} = 2q - 3, \quad P_{k+1,t} = (2tu + 5 - 4q)P_{k,t} + \partial_u P_{k,t} + 4q\partial_q P_{k,t}.$$

For $0 \le t \le 1$, define nonnegative-coefficient polynomials

$$Q_0 = 2q + 3, \quad Q_{k+1} = (2u + 5 + 4q)Q_k + \partial_u Q_k + 4q\partial_q Q_k.$$

Coefficientwise absolute-value comparison in the polynomial variables t, u, q, followed by $0 \le t \le 1$, proves $|P_{k,t}(u, q)| \le Q_k(u, q)$ for $u, q \ge 0$. Leibniz's rule gives the positive polynomial

$$W_{Kj}(u, q) = \sum_{\ell=0}^{\min(j,2K)} \binom{2K}{\ell} \frac{j!}{(j-\ell)!} u^{j-\ell} Q_{2K-\ell}(u, q) = \sum_{a,b} \omega_{ab} u^a q^b.$$

All ω_{ab} are nonnegative. Also write $Q_{2k+1}(0, q) = \sum_b \eta_{kb} q^b$, and put

$$B_{Kjk} = \left\| (u^j \beta_k)^{(2K)} \right\|_{L^1(0,\infty)}.$$

These are explicit finite Gaussian-polynomial integrals. An elementary upper bound is obtained by summing absolute polynomial coefficients times $\int_0^\infty u^a e^{-u^2}\,du = \Gamma((a+1)/2)/2$.

Choose N so that $\alpha_{N+1} = \pi(N+1)^2 \geq 8K+9$. For every integer s occurring below, define

$$T_s(N) = \frac{\alpha_{N+1}^s e^{-\alpha_{N+1}}}{1-e^{-\pi(N+2)}}.$$

Then the complete all-index estimate is

$$\left\|(u^j R_N)^{(2K)}\right\|_1 \leq \sum_{a,b} \omega_{ab} \frac{a!}{3^{a+1}} T_{b-a}(N) + \sum_{k=0}^{K-1} B_{Kjk} \sum_b \eta_{kb} T_{b+1}(N). \tag{14.4}$$

To prove this, a monomial $u^a q^b$ in the derivative bound contributes

$$\alpha_n^{b+1} \int_0^\infty u^a e^{u^2+(5+4b)u-\alpha_n e^{4u}}\,du \leq \frac{a!}{3^{a+1}} \alpha_n^{b-a} e^{-\alpha_n}.$$

Here $b \leq 2K+1$, so $5+4b \leq 8K+9 \leq \alpha_n$. The inequality $e^{4u} \geq 1+4u+8u^2$ leaves a linear decay rate at least $3\alpha_n$ and a nonpositive quadratic exponent. The endpoint coefficients obey

$$|g_{n,t}^{(2k+1)}(0)| \leq e^{-\alpha_n} \sum_b \eta_{kb} \alpha_n^{b+1}.$$

Finally every positive exponent s in these bounds satisfies $s \leq 2K+1$, hence $2s \leq \alpha_n$. For $n \geq N+1$, the successive ratio of $\alpha_n^s e^{-\alpha_n}$ is at most $e^{-\pi(n+1)}$: use $\log(1+1/n) \leq 1/n$ if $s > 0$; for $s \leq 0$ the bound is even smaller. Summing the whole geometric majorant gives $\sum_{n>N} \alpha_n^s e^{-\alpha_n} \leq T_s(N)$. Applying the triangle inequality to the compensated source proves (14.4).

Compensation is a decomposition at each fixed time. Its separate channels need not solve heat or be positive. Finite K improves a bounded-height certificate by paying known endpoint cancellations; it still gives algebraic error, not a global error relative to the potentially much smaller physical norm.

14.1 The explicit two-jet compensation

For $K = 2$, choose

$$\beta_0 = (u+u^3)e^{-u^2}, \qquad \beta_1 = u^3 e^{-u^2}/6, \qquad b_n = \phi_n'(0), \quad e_n = \phi_n'''(0).$$

Their first and third endpoint jets are respectively $(1,0)$ and $(0,1)$. Thus $\widetilde{g}_{n,t} = g_{n,t} - b_n\beta_0 - (e_n + 6tb_n)\beta_1$. Write $G_n = \int_0^\infty g_{n,t}\cos(xu)\,du$, $B_k = \int_0^\infty \beta_k \cos(xu)\,du$, and $a_n = G_n - b_n B_0 - (e_n + 6tb_n)B_1$. The exact residual is

$$(\partial_t + \partial_x^2)a_n = -b_n B_0'' - (e_n + 6tb_n)B_1'' - 6b_n B_1. \tag{14.5}$$

The raw channel G_n obeys heat. Its compensation is carried explicitly as an affine function of time; treating a_n as a homogeneous heat solution would omit (14.5).

For $N = 3$, the complete-index payments from (14.4) can be made rational. If $A = 16\pi$, then for every integer exponent in that formula,

$$T_s(3) < \frac{(3/8)^{50}}{1-(3/8)^{15}} \begin{cases} (352/7)^s, & s \geq 0, \\ 50^s, & s < 0. \end{cases}$$

Here $25/8 < \pi < 22/7$ and $e > 8/3$ suffice. Four Gaussian-polynomial differentiations give upper bounds for B_{2jk}, with rows $j = 0, 1, 2, 3$ and columns $k = 0, 1$:

$$\begin{pmatrix} 316 & 67 \\ 702 & 747/5 \\ 1566 & 328 \\ 18819/5 & 1539/2 \end{pmatrix}.$$

Insert these into the positive-polynomial formula, using

$$Q_1(0,q) = 15 + 30q + 8q^2, \quad Q_3(0,q) = 465 + 3450q + 4280q^2 + 1440q^3 + 128q^4.$$

If $P_3 = \sum_{n\le 3} a_n$ and $r_k = h_k - \partial_x^k P_3$, the resulting complete remainders satisfy, uniformly for $0 \le t \le 1$,

$$|r_k| < c_k |x|^{-4}, \qquad (c_0, c_1, c_2, c_3) = (2, 4, 9, 20)10^{-9}. \tag{14.6}$$

The exact unrounded rational comparisons also give $(|r_0|, |r_1|, |r_2|, |r_3|) < (1, 2, 4, 9)10^{-12}$ for $|x| \ge 7$. These are index-tail payments; retained quadrature and parameter approximation errors require separate enclosures.

14.2 The correlated stationary observation

For $a_{n,k} = \partial_x^k a_n$, $p_k = \sum_{n\le N} a_{n,k}$, put

$$D_j^N = \sum_{n\le N} (a_{n,j+1}^2 - a_{n,j} a_{n,j+2}),$$
$$X_j^N = \sum_{n<m\le N} (2a_{n,j+1}a_{m,j+1} - a_{n,j}a_{m,j+2} - a_{m,j}a_{n,j+2}).$$

Then $D_j^N + X_j^N = p_{j+1}^2 - p_j p_{j+2}$ exactly. At genuine stationarity, $h_{j+1} = 0$, substitution of $p_{j+1} = -r_{j+1}$ gives

$$\mathscr{L}_j(H) = D_j^N + X_j^N - r_{j+1}^2 - p_j r_{j+2} - p_{j+2} r_j - r_j r_{j+2}. \tag{14.7}$$

Thus for $|r_k| \le \varepsilon_k$, the payment is

$$E_j = \varepsilon_{j+1}^2 + |p_j|\varepsilon_{j+2} + |p_{j+2}|\varepsilon_j + \varepsilon_j\varepsilon_{j+2}. \tag{14.8}$$

A lower enclosure $D_j^N + X_j^N - E_j > \mu_j > 0$ on a cell proves $\mathscr{L}_j(H) > \mu_j$ wherever that cell meets the genuine stationary set. Aggregate evaluation can preserve correlation, while the exact identity retains every off-diagonal term. Neither D_j^N nor X_j^N is assigned an individual sign.

For example, on $[1/200, 3/50] \times [7, 8]$, (11.13) gives $\mathscr{L}_1(H) > 49/240000000000$. With the uniform four-jet index errors above and the genuine jet caps $(1/2, 1/8, 1/12, 1/12)$,

$$E_1 < 13\,10^{-13}, \qquad |\mathscr{L}_1(H) - \mathscr{L}_1(P_3)| \le 2|p_2|\varepsilon_2 + E_1 < 2\,10^{-12}.$$

Consequently $X_1^3 > -D_1^3 + E_1 + 10^{-10}$ throughout that band. This analytic cross-term bound does not require stationarity there.

14.3 Guarded forward continuation of stationary signs

Proposition 14.2 (A paid stationary time collar)**.** *Fix $j \in \{0,1\}$, a spatial interval $[a,b]$, and $0 \le t_0 < t_0 + \Delta \le \tau \le 1$. Suppose at time t_0,*

$$\inf_{[a,b]} \sqrt{h_{j+1}^2 + h_{j+2}^2} \ge d_j > 0, \qquad \min_{x=a,b} |h_{j+1}| \ge e_j > 0.$$

Assume $-h_j h_{j+2} > 0$ at every stationary root on that slice. Set $B_j = \sqrt{M_{j+3}(\tau)^2 + M_{j+4}(\tau)^2}$. If

$$\Delta B_j < d_j, \qquad \Delta M_{j+3}(\tau) < e_j, \tag{14.9}$$

then every stationary root throughout the closed slab lies on a unique regular continuation of an initial root, remains in (a,b), and satisfies $\mathscr{L}_j > 0$. On a branch with initial $a_\nu = |h_j(t_0, x_\nu)|$,

$$\mathscr{L}_j(t, \chi_\nu(t)) \ge a_\nu[d_j - (t - t_0)B_j] > 0, \qquad \mathcal{E}(t, \chi_\nu(t)) \ge a_\nu^2.$$

Proof. Source positivity gives $|h_m(t,x)| \le M_m(\tau)$. The heat equation therefore bounds the change of the derivative pair in time by $(t - t_0)B_j$, and its norm stays at least $d_j - (t - t_0)B_j > 0$. The edge derivative retains its strict sign by the second guard. Consequently stationary roots neither become degenerate nor cross the vertical edges. The implicit function theorem and compactness give complete branch continuation forward and backward inside the slab; in particular no stationary pair can appear unseen. There are finitely many initial roots, since they are regular and the edges are nonzero.

On such a branch put $A = h_j$, $C = h_{j+2}$. Differentiation gives

$$\chi_\nu' = h_{j+3}/C, \qquad A' = -C.$$

The nonzero C has constant sign, opposite the initial sign of A. Thus $|A|$ increases forward and is at least a_ν. At stationarity the derivative-pair bound is a lower bound on $|C|$. Multiplying proves the stated signed margin. Energy contains A^2 for either $j = 0$ or $j = 1$, which proves its bound without division by an unproved physical norm. □

All moments in these guards include the full theta source. For example, for $m \ge 0$, $0 \le \tau \le 1$,

$$M_m(\tau) \le 2m! \sum_{n \ge 1} \frac{\alpha_n^2 e^{-\alpha_n}}{(4\alpha_n - 9)^{m+1}}.$$

The tail beyond N is bounded by its first omitted term divided by $1 - \rho_N$, where $\rho_N = ((N+2)/(N+1))^4 e^{-\pi(2N+3)} < 1$. The guards remain absolute: they can demand very short time steps at large heights, and require independent slice and edge certificates.

They cannot be omitted. With $y = x - a$, $\delta = t - T$, $A_* > 0$, the backward-heat control

$$Q = y^4/4 - 3\delta y^2 + 3\delta^2 - A_*$$

has a positive first stationary sign and an empty second stationary set at $\delta = -h$, $0 < h < \sqrt{A_*/3}$. At $\delta = 0$ its stationary derivative is $Q_x = y^3$, so the derivative-pair gap closes. For small $\delta > 0$, the central stationary sign becomes $\mathscr{L}_0(t,a) = -6\delta(A_* - 3\delta^2) < 0$, even though $Q(T,a) = -A_* \ne 0$. Degeneracy of a stationary branch must therefore be checked before any physical collision, not inferred absent from its initial sign.

14.4 A certificate over the full threshold time range

Theorem 14.3 (Validated signs and norm through height 41)**.** *For the complete theta family, on the closed domain* $0 \le t \le 1/2$, $7 \le |x| \le 41$,

$$h_1 = 0 \Longrightarrow \mathscr{L}_0 > 5\,10^{-10}, \qquad h_2 = 0 \Longrightarrow \mathscr{L}_1 > 10^{-11}. \tag{14.10}$$

On the entire central strip at these times,

$$|F(t,x)| > \frac{9}{5000000}, \qquad 0 \le t \le 1/2, \quad |x| \le 41. \tag{14.11}$$

Both stationary signs hold on this entire strip. Consequently any positive threshold collision, if one exists, has $|a| > 41$.

Validated calculation. The reproducible implementation uses two Python sources:

`certify_compensated_stationary_time_extension.py`
`certify_compensated_stationary_theta.py`.

The second supplies the retained-channel machinery imported by the first. The small execution record binds both source hashes. The driver covers $[0,1/2] \times [7,41]$ by the 272 closed cells

$$[r/16,(r+1)/16] \times [k,k+1], \qquad 0 \le r < 8, \quad 7 \le k < 41.$$

All endpoints and the gap-free cover are checked with exact fractions. The run uses $K = 2$, $N = 3$, 256-bit Arb/Acb arithmetic, and a retained-source integration cutoff $U = 1$.

Here are the complete parameter and integration payments. For each time slab use its midpoint t_c, radius $d = 1/32$, maximum t_+, spatial midpoint x_c, and radius $\rho = 1/2$. With $J_{n,k} = \partial_x^k G_n(t_c, x_c)$, the raw-channel polynomial is

$$T_{n,k} = \sum_{\ell=0}^{4} \sum_{m=0}^{11} \frac{(-(t-t_c))^\ell (x-x_c)^m}{\ell! m!} J_{n,k+2\ell+m}.$$

Writing $M_{n,q}(t_+) = \int_0^\infty u^q g_{n,t_+}(u)\,du$, its remainder is at most

$$\sum_{\ell=0}^{4} \frac{d^\ell}{\ell!} \frac{\rho^{12}}{12!} M_{n,k+2\ell+12}(t_+) + \frac{d^5}{5!} M_{n,k+10}(t_+).$$

This is Taylor's theorem first in time and then in space, applied to the raw heat solution, with positive source moments at the slab maximum. Validated Acb quadrature encloses its returned error on $[0,1]$. For the entire remaining half-line put $L_n = t_+ + 9 - \alpha_n e^4$, $\lambda_n = 4\alpha_n e^4 - 9 - 2t_+ > 0$. Concavity gives the tail bound

$$2\alpha_n^2 e^{L_n} \sum_{\ell=0}^{q} \binom{q}{\ell} \frac{\ell!}{\lambda_n^{\ell+1}}.$$

The same payment is included in every moment quadrature.

The Gaussian compensation is evaluated on the whole half-line by

$$I_0 = \tfrac{\sqrt{\pi}}{2} e^{-x^2/4}(1 + i\,\mathrm{erfi}(x/2)), \quad I_1 = (1 + ixI_0)/2, \quad I_{m+1} = (mI_{m-1} + ixI_m)/2,$$

where $I_m = \int_0^\infty u^m e^{-u^2+ixu}\,du$. Then $B_0^{(k)} = \mathrm{Re}[i^k(I_{k+1}+I_{k+3})]$, $B_1^{(k)} = \mathrm{Re}[i^k I_{k+3}/6]$. Their spatial Taylor polynomials through degree 23 have remainders $\rho^{24}(\mu_{k+25}+\mu_{k+27})/24!$ and $\rho^{24}\mu_{k+27}/(6\cdot 24!)$,

respectively, with $\mu_q = \Gamma((q+1)/2)/2$. Explicit affine-time compensation retains (14.5). The complete-index bound (14.4) is valid uniformly for $0 \le t \le 1$, so every omitted index is paid on each slab.

At level zero, stationarity is excluded in 263 cells and a signed surplus is certified in nine. At level one the corresponding counts are 255 and seventeen. The smallest exact lower surpluses exceed $5.44\,10^{-10}$ and $1.44\,10^{-11}$, respectively; their outward dyadic endpoints decide the rational targets in (14.10) through (14.7)–(14.8). Every exclusion is a genuine derivative separation or a uniform endpoint-sign test with strict spatial monotonicity. Adjacent signed cells are joined when a branch crosses a cell boundary. Strict next derivative signs and opposite outer endpoint signs isolate one first and two second stationary branches at each time. No sign cell or branch window remains unresolved.

There is a separate direct norm payment. On a cell let J_0, J_1 be its full genuine jet intervals, including all errors above, and put $v_k = \operatorname{dist}(0, J_k)$. Exact rational evaluation gives

$$H^2 + H_x^2 \ge v_0^2 + v_1^2 > \left(\frac{9}{5000000}\right)^2$$

on every cell. The certified global norm lower bound is approximately $1.89166\,10^{-6}$; the rational comparison, rather than this decimal display, establishes (14.11). Evenness covers negative heights. The central norm gap and Proposition 11.3 cover $|x| \le 8$, with overlap. The threshold criterion restricts a positive threshold to $0 < t \le 1/2$; the direct norm bound therefore excludes its collision through height 41, including the boundaries. □

14.5 Correlated transport through height 100

Theorem 14.4 (Higher-height validated cover)**.** *On $0 \le t \le 1/2$, $40 \le |x| \le 100$,*

$$h_1 = 0 \Longrightarrow \mathscr{L}_0 > 10^{-30}, \qquad h_2 = 0 \Longrightarrow \mathscr{L}_1 > 2\,10^{-32}.$$

Moreover,

$$|F(t,x)| > 10^{-16} \qquad (0 \le t \le 1/2,\ |x| \le 100). \tag{14.12}$$

Both stationary signs hold on this whole strip. Any positive threshold collision must therefore have $|a| > 100$.

Validated calculation. The new driver evaluates the aggregate raw-source Taylor polynomial by first summing center jets over theta indices:

$$\sum_{n\le N}\sum_{\ell<L}\sum_{m<M}\frac{(-\tau)^\ell\eta^m}{\ell!m!}J_{n,k+2\ell+m} = \sum_{\ell<L}\sum_{m<M}\frac{(-\tau)^\ell\eta^m}{\ell!m!}\sum_{n\le N}J_{n,k+2\ell+m}.$$

This exact reordering retains cancellation in common coefficients. It still adds every center-jet radius and every positive per-channel Taylor remainder, before applying the summed affine-time compensation. Individual channels and all diagonal and cross terms remain enclosed. The same physical observation is computed; no error payment is removed.

The run uses $K = 2, N = 4$, 256-bit Arb/Acb arithmetic, cutoff $U = 1$, space order 16, time order 8, and Gaussian order 32. Its eight closed time slabs are $[r/16, (r+1)/16]$, $0 \le r < 8$. The three closed spatial bands $[40, 60], [60, 80], [80, 100]$ use 160, 170, and 180 cells, respectively. Unit spatial cells are bisected where signs or a direct norm remain unresolved, to depth at most two; no time bisection is used. Exact two-dimensional cover checks include every shared and outer

boundary. All 510 final cells have both stationary observations resolved, a positive direct norm floor, and resolved merged branch windows.

The complete-index, retained integration, Gaussian, quadrature, rounding, and parameter payments are those above with the new declared orders and N. The general finite-pair identity retains every ordered index pair at $N = 4$. The smallest exact lower stationary surpluses exceed $1.8018\,10^{-30}$ and $2.1514\,10^{-32}$, respectively. The genuine-jet interval-distance calculation gives the norm floor

$$\frac{22271745300542300149807}{85070591730234615865843651857942052864} > 10^{-16}.$$

The minimizing cell is $[1/16, 1/8] \times [96, 97]$. This is a direct lower bound from paid observations. Evenness and the overlap with Theorem 14.3 prove (14.12). The three compact records bind the new driver and imported source; independent internal replay audits their exact decisions. They are finite-height certificates, not an all-height coverage argument. □

15 Shifted contours and the remaining moving-domain target

Proposition 15.1 (Complete-source shifted physical jets)**.** *For $0 \le t \le \tau \le 1/2$, $x \ge 0$, $0 < a < \pi/8$, and every integer $k \ge 0$,*

$$h_k(t,x) = e^{-ax}\,\mathrm{Re}\int_0^\infty (i(u+ia))^k e^{t(u+ia)^2}\Phi(u+ia)e^{ixu}\,du. \tag{15.1}$$

This identity is first established for the complete theta source.

Proof. The theta series is normally convergent on compact subsets of $|\mathrm{Im}\, z| < \pi/8$, because $\mathrm{Re}\, e^{4z} > 0$. Analytic continuation of Jacobi reflection gives $\Phi(-z) = \Phi(z)$ there. Schwarz reflection then makes $\Phi(iy)$ real for $|y| < \pi/8$. Shift the half-line Fourier integral to height a. On its left vertical segment $z = iy$, all factors of $(iz)^k e^{tz^2}\Phi(z)e^{ixz}$ are real, whereas $dz = i\,dy$. This segment therefore has zero real part for every physical jet. On the right edge, $\cos(4y) \ge \cos(4a) > 0$ ensures uniform double-exponential decay. Its integral tends to zero. Cauchy's theorem proves (15.1). Individual theta summands do not have this vertical cancellation; truncation follows the complete identity, with the omitted horizontal source paid below. □

Proposition 15.2 (Explicit horizontal truncation payments)**.** *Put $c_a = \cos(4a) > 0$, $m = N + 1$, $\alpha = \pi m^2$, and*

$$\beta = c_a\alpha, \quad \lambda = 4\beta - 9, \quad \rho = ((m+1)/m)^4 e^{-\pi c_a(2m+1)}, \quad S_k(\lambda, b) = \sum_{r=0}^{k}\binom{k}{r}\frac{b^{k-r}r!}{\lambda^{r+1}}.$$

If $\rho < 1$, $\lambda > 0$, $8\beta \ge \tau$, the complete omitted-index error in (15.1) is at most

$$E_{N,k} = e^{-ax}\frac{(2\alpha^2 + 3\alpha)e^{-\beta}}{1-\rho}S_k(\lambda, a). \tag{15.2}$$

For an integration cutoff $U \ge 0$, put

$$b_U = \pi c_a e^{4U}, \quad \rho_U = 16e^{-3b_U}, \quad \lambda_U = 4b_U - 9 - 2\tau U, \quad L_U = \tau U^2 + 9U - b_U.$$

If $\rho_U < 1$, $\lambda_U > 0$, $8b_U \ge \tau$, the entire source on $u \ge U$ contributes at most

$$E_{U,k} = e^{-ax}\frac{(2\pi^2 + 3\pi)e^{L_U}}{1-\rho_U}S_k(\lambda_U, U + a). \tag{15.3}$$

Thus the finite N-term integral on $[0, U]$ has total error at most $E_{N,k} + E_{U,k}$, before adding validated retained quadrature and parameter errors.

Proof. On the shifted half-line,

$$|\phi_n(u+ia)| \le (2\alpha_n^2+3\alpha_n)e^{9u-c_a\alpha_n e^{4u}}, \quad |e^{t(u+ia)^2}| \le e^{\tau u^2}, \quad |u+ia|^k \le (u+a)^k.$$

For indices $n \ge m$, the successive ratios of the two coefficient majorants are at most ρ, uniformly in $u \ge 0$. The lower bound $e^{4u} \ge 1+4u+8u^2$ gives

$$\tau u^2+9u-\beta e^{4u} \le -\beta-\lambda u-(8\beta-\tau)u^2 \le -\beta-\lambda u.$$

Integrate $(u+a)^k$ against this exponential after summing the complete geometric index bound to obtain (15.2). For $u \ge U$, the corresponding index ratio is at most ρ_U. The exponent $L(u) = \tau u^2+9u-\pi c_a e^{4u}$ is concave there and has slope $-\lambda_U$ at U. Its tangent bound proves (15.3). Adding these two payments overcounts their intersection, which is harmless for an upper bound. □

Corollary 15.3 (Absolute errors at the gamma exponential scale)**.** *For $x \ge 41$, fixed $p > 0$, and $\ell = \log x$, choose*

$$a_x = \pi/8-1/x, \quad m = \left\lceil \sqrt{\frac{x}{2\pi}(p\ell+32)} \right\rceil, \quad N = m-1, \quad U = \frac{\ell+2\log(\ell+8)}{4}.$$

All hypotheses of Proposition 15.2 hold. For each fixed k, uniformly in $0 \le t \le 1/2$,

$$e^{\pi x/8}E_{N,k} = O_{p,k}(x^{5/2-p}(\log x)^{3/2}), \qquad e^{\pi x/8}E_{U,k} = O_{k,A}(x^{-A}) \quad (A>0). \tag{15.4}$$

These are absolute error estimates, with no sign or lower norm assertion.

Proof. Here $c_a = \sin(4/x) \ge 2/x$, $\beta \ge p\ell+32$, and $m^2 \ge 2x/\pi$. Therefore

$$-\log\rho \ge 4\pi m/x-4/m \ge 2\pi m/x, \qquad (1-\rho)^{-1} \le 1+x/(2\pi m).$$

Also $e^{-a_x x} = e\,e^{-\pi x/8}$. Substitution in (15.2) proves its stated order, including the growing geometric denominator. For the integration tail, $b_U \ge 2\pi(\ell+8)^2$ and $U \le \ell/2+2$. Hence

$$L_U \le -47\ell^2/8-181\ell/2-364.$$

The remaining factors in (15.3) grow at most polynomially in ℓ, proving the second estimate. □

On a closed spatial cell $[X,Y]$ with $Y \ge 41$, freeze $a = a_Y$, $N = N(Y)$, and $U = U(Y)$. Then $e^{(\pi/8-a_Y)x} \le e$ throughout the cell, giving uniform normalized truncation payments. Every physical jet is formed by (15.1) before choosing the contour; differentiating a finite approximation with variable shift or cutoffs would introduce additional terms. Retained quadrature and whole-cell parameter variation remain separate obligations. No all-height cell calculation is asserted by (15.4).

15.1 The moving small-time region and its signed target

The imported uniform localization theorem [4, Theorem 1.5(i)] has a sufficiently large absolute constant C: a nonreal zero at $0 < t \le 1/2$ has $|\operatorname{Re} z| < e^{C/t}$. At a multiple real zero (T,a) with $0 < T \le 1/2$, local Hermite splitting produces nonreal zeros at $T-h$ with real parts tending to a. Taking the limit in that localization bound gives

$$T\log|a| \le C \qquad (|a|>1). \tag{15.5}$$

Strictness is lost in the limit. No explicit numerical C is imported. A padded moving domain uses $t\log|x| < C'$ with $C' > C$, which contains every sufficiently close predecessor of a candidate collision. Fixed-positive-time asymptotics do not alone cover its unbounded-height limit as $t \downarrow 0$.

Put $c = \pi/8$ and define normalized physical jets $Z_k = e^{cx}h_k$, $\mathcal{N} = Z_0^2 + Z_1^2$. Then

$$\partial_x Z_k = Z_{k+1} + cZ_k, \quad \partial_t Z_k = -Z_{k+2}, \quad \mathcal{Q}_j = Z_{j+1}^2 - Z_j Z_{j+2} = e^{2cx}\mathscr{L}_j.$$

Stationarity is $Z_{j+1} = 0$, not $\partial_x Z_j = 0$. On a moving genuine stationary branch,

$$\tfrac{1}{2}\frac{d}{dt}Z_j(t,\chi(t))^2 = \mathcal{Q}_j + c\chi'(t)Z_j(t,\chi(t))^2.$$

Thus normalized amplitude growth does not by itself give the physical signed observation. For paid normalized approximations $Z_k = p_k + r_k$, $|r_k| \le \varepsilon_k$, genuine stationarity yields

$$\mathcal{Q}_j = p_{j+1}^2 - p_j p_{j+2} - r_{j+1}^2 - p_j r_{j+2} - p_{j+2}r_j - r_j r_{j+2}.$$

The stationary payment (14.8) therefore applies verbatim to the normalized jets. Proving a nonnegative surplus, or a loss locally bounded by $\mathcal{N}$, throughout the padded moving domain remains the arithmetic obligation. Absolute error tending to zero does not establish that surplus. The criterion requires positive time; it does not impose simplicity of all time-zero scalar zeros.

16 What remains to prove

The quaternion field is fixed by its theta trace, its jet constraint, and its heat-time displacement. Its regularity and its Cauchy formula are exact identities. The bounded central estimate proves nonvanishing on its declared rectangle, and the gluing theorem quantitatively controls errors in a declared coefficient class. The validated extension now covers the full possible positive-threshold time range through height 100. Any positive threshold collision must therefore occur at $|a| > 100$. No estimate here supplies a collision-excluding sign at every height.

A sufficient next result would bound the two signed physical observations on predecessor neighborhoods by a loss locally bounded through joint zeros. Equivalently, one may control the time derivative of the physical norm on the union of the two stationary sets. The boundary expressions give the same signed observations as full theta-index pairings. Their cross terms and endpoint cancellations must be retained. The scalar whole-plane curvature criterion provides a separate exact route, with the scalar projection of the quaternion trace as its observation.

These statements specify several equivalent or sufficient mathematical targets. They do not promote Fueter regularity, a subharmonic quaternion norm, or an absolute boundary derivative estimate to an arithmetic sign theorem.

Proposition 14.2 reduces a forward step to complete-source derivative-pair and edge gaps plus an initial signed amplitude. At increasing heights those gaps can be much smaller than the absolute moment constants. A moving-domain program must prove every step's regularity, entry and creation guards, and full predecessor coverage. Fixed-order algebraic compensation errors cannot be divided by a potentially much smaller unproved physical norm to obtain a global relative estimate. The shifted-contour family removes that particular algebraic approximation ceiling, but leaves the normalized stationary surplus unproved in $t\log|x| < C'$. This moving small-time region, with its genuine stationary constraint, is the next signed target.

A Reproducible algebra and constant checks

All analytic arguments used for the stated theorems are in the text. The following small standard-library Python checks independently replay polynomial identities and exact constants; they do not decide an unproved theta sign by sampling.

The file `check_fueter_boundary_identities.py` tests polynomial backward-heat solutions of degrees $0, 1, 2, 3, 4, 6$, and the quadratic control, at two rational choices of time, center, and radius. Its 14 cases check the real-coordinate Fueter residual, the first three normalized spherical moments, and the signed boundary correlations. Additional controls check a positive-time joint zero and a negative stationary sign. Integration of polynomial sphere monomials uses

$$\langle \omega_0^{2a_0} \cdots \omega_3^{2a_3} \rangle = \frac{\prod_{\mu=0}^{3} (2a_\mu - 1)!!}{4 \cdot 6 \cdots (4 + 2\sum a_\mu - 2)},$$

with the empty product equal to one; a monomial with an odd exponent has zero average. Thus no numerical sphere quadrature enters these checks.

The file `check_theta_central_moment_bound.py` verifies 17 exact rational comparisons for the complete-source core and tail, the signed moment margin, and its consequences. The file `check_theta_gluing_defect_constants.py` verifies 22 comparisons or polynomial identities for the four jet constants, the signed-form error bounds, the robust central margins, and the remote-control determinant. Exponential comparisons follow from finite positive Taylor sums or a geometric bound on their remaining terms. A rational interval for π in the determinant check follows from Machin's identity and alternating arctangent series. The 39 decisions use exact fractions.

The scripts and their small execution records are in the investigation's `numerics/` directory. Each accepts `-output` followed by a JSON path. The records bind the source scripts by SHA-256 and record the arithmetic and scope. Matching hashes establish file identity; the displayed analytic proofs establish the mathematical implications. The additional file `check_quaternion_coordinate_algebra.py` checks the Hamilton pair product and conjugation, complex lift values, standard Fueter and external heat identities, signed complex Laguerre components, and the real Jacobian orientation at rational complex coordinate values. All 58 exact checks pass. It records both its own source hash and that of the polynomial utility it imports.

The additional `check_compensated_tail_and_signed_overlap.py` replays 46 exact rational comparisons for the two-jet index payments and the fourth–sixth moment gap. The full-time extension's compact record `COMPENSATED_STATIONARY_TIME_EXTENSION_20261014.json` contains the 272 closed cells, their paid genuine jets, exact cover checks, signed decisions, merged branch windows, and direct norm lower bound. Its independent replay agrees except for elapsed runtime. The file `check_forward_stationary_collars.py` supplies 234 exact checks, including polynomial branch identities, the quartic control, and a concrete short theta time collar derived from the older certificate. The guarded continuation proof remains the analytic argument in Proposition 14.2.

The higher-height driver is

`certify_correlated_source_height_extension.py`.

Its three records, with prefix `CORRELATED_SOURCE_HEIGHT`, cover the bands 40–60, 60–80, and 80–100. Independent replay agrees except for elapsed runtime; 6682 exact semantic and rational audit checks verify the cover, general four-index pair identity, stationary decisions, branch windows, source bindings, and direct norm extraction. The shifted-contour checker verifies the payment hypotheses at five selected heights from 41 to 10^{12}, and compares eight paid shifted and unshifted jet enclosures at height 41. Its large-height checks evaluate the analytic tail formulas, not millions of theta terms or an all-height sign. The universal contour and cusp arguments are the analytic proofs in Section 15.

A.1 Exact constants and their elementary bounds

The accompanying standard-library Python checkers use exact rational arithmetic for every decision. Their two execution records contain 39 passing checks, together with the SHA-256 hash of each checker source. Seventeen checks support the complete-source central theorem: finite exponential-series bounds for $e < 11/4$, $e > 8/3$, $e^{2/5} < 3/2$, $e^{28/25} > 153/50$; the complete-index geometric ratio; the core constant; the tail exponent and slope; the $e^{69/10} > 800$ payment; the signed tail integral and moment margin; and the scalar, slope, curvature and origin stationary consequences. For $x > 0$, a finite lower exponential sum is $\sum_{k=0}^{L} x^k/k!$. When $x < L + 2$, a valid upper sum adds

$$\frac{x^{L+1}}{(L+1)!}\frac{1}{1 - x/(L+2)}.$$

These inequalities supply the finite rational comparisons used above. The classical rational bounds on π are $25/8 < \pi < 22/7$ and $\pi^2 < 10$.

The remaining 22 checks support the gluing theorem and control: $e^3 > 20$; all four squared constant comparisons; the linear and quadratic coefficients in the two signed-form transfers; the robust slope, curvature, inner-value and norm gaps; the origin stationary margin; the symbolic recurrences for P_1, P_2, P_3; the determinant polynomial; and its strict sign. For the last sign, Machin's identity

$$\pi = 16\arctan(1/5) - 4\arctan(1/239)$$

and alternating rational sums (12 terms and 4 terms, respectively) enclose

$$\frac{314159265358979}{10^{14}} < \pi < \frac{314159265358980}{10^{14}}.$$

Rational interval Horner evaluation on this interval gives $4598266/10^6 < P(\pi) < 4598268/10^6$. No numerical quadrature, sampled Fourier sign, finite-source substitution, or zeta-zero table enters any sign decision. The checks verify the stated finite comparisons; the analytic arguments above supply the domain, convergence, complete-tail and source-class implications. These are reproducibility checks with internal LLM-assisted review, not independent specialist validation or a global Riemann-hypothesis certificate.

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This manuscript was drafted for Edward Baker with substantial assistance from OpenAI's Codex using the recorded model `gpt-6.1-sol` at reasoning effort `ultra`, including parallel LLM checks of the RH foundation, quaternion conventions, and theta estimates. The mathematical development consolidates investigation 21's research notes of 14 October 2026 and the collision framework developed in investigation 20. The checks described here are internal verification, not independent specialist refereeing or machine-checked formal proofs.

References

[1] NIST Digital Library of Mathematical Functions, *Theta functions: Mellin transforms*, Equation 20.10.2. https://dlmf.nist.gov/20.10.E2.

[2] N. G. de Bruijn, *The roots of trigonometric integrals*, Duke Mathematical Journal **17** (1950), 197–226. https://doi.org/10.1215/S0012-7094-50-01720-0.

[3] C. M. Newman, *Fourier transforms with only real zeros*, Proceedings of the American Mathematical Society **61** (1976), 245–251. https://doi.org/10.1090/S0002-9939-1976-0434982-5.

[4] D. H. J. Polymath, *Effective approximation of heat flow evolution of the Riemann ξ function, and a new upper bound for the de Bruijn–Newman constant*, arXiv:1904.12438v2 (2019). https://arxiv.org/abs/1904.12438.

[5] C. M. Newman and W. Wu, *Constants of de Bruijn–Newman type in analytic number theory and statistical physics*, arXiv:1901.06596 (2019). https://arxiv.org/abs/1901.06596.

[6] B. Rodgers and T. Tao, *The de Bruijn–Newman constant is non-negative*, Forum of Mathematics, Pi **8** (2020), e6. https://arxiv.org/abs/1801.05914.

[7] A. Sudbery, *Quaternionic analysis*, Mathematical Proceedings of the Cambridge Philosophical Society **85** (1979). https://doi.org/10.1017/S0305004100055638.

[8] I. Frenkel and M. Libine, *Quaternionic analysis, representation theory and physics*, Advances in Mathematics **218** (2008), 1806–1877. https://arxiv.org/abs/0711.2699.

[9] G. Csordas and A. Escassut, *The Laguerre inequality and the distribution of zeros of entire functions*, Annales mathematiques Blaise Pascal **12** (2005), 331–345. https://www.numdam.org/item/AMBP_2005__12_2_331_0/.